

Experimental apparatus for universal spin-motion control in multi-ion registers



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k-copy Randomised Benchmarking

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Check framing carefully.

To construct useful scale quantum processors we must house many qubits physically close together, and must ensure the full register operates with “low error rates”.

But how should we define our errors in a large system? Naively, one might hope that the error rate of a multi-qubit system could be inferred from the error rates of its constituent qubits, but unfortunately, not all errors are commensurable — there are a zoo of possible error channels. They may be coherent, incoherent, continuous, stochastic, local, correlated, time dependent or independent (and I am sure a host of further adjectives exist). In response, a zoo of error correction and mitigation techniques have been developed each with their own scaling overheads and assumptions on these noise models they treat [1–3]. It is a large and intimidating search space, and so, to better equip ourselves for navigation, we require tools to characterise what errors are relevant in our physical systems. Crucially, errors which induce correlations across multiple qubits, termed spatial correlations or crosstalk errors [4, 5], require mitigation and correction strategies built on noise models that faithfully represent their correlated structure [6]; assuming globally depolarising noise models (white noise), can lead to systematic underestimation of logical error rates [7, 8], or systematic bias of the corrected observable outcome [9]. In this chapter we introduce a scalable method for characterising correlated errors in multi-qubit systems: *k*-copy randomised benchmarking. This technique extends single-qubit Clifford RB [10–12] (discussed in chapter ??) to parallel application across multiple qubits, where now symmetry resolved dynamics provides evidence of, and quantifies various multi-qubit error channels.

This chapter is structured as follows: we start by giving an overview on the problem of *characterisation*, and what techniques already exist to address it. We then outline the *k*-copy protocol and discuss the relevant symmetric subspace structure. We derive the population dynamics when subject to simple noise models and discuss variations of the protocol that isolate specific subspaces. We provide experimental demonstrations of the protocol on a trapped-ion processor through injecting known correlated and uncorrelated errors. We end this chapter by demonstrating the scalability of the protocol on an 8-qubit register, and discuss the limitations and potential of this method for further scaling to larger systems.

1.1 Errors and Characterisation

Errors, which we shall give a broad definition as any deviations between the realised and ideal state of a quantum information processor (QIP), are due to unintentional environmental coupling and control imperfections. They are unavoidable, and if not carefully considered or mitigated, lead to unreliable or unverified computation. The process of modelling and quantifying these errors is known as *characterisation*. Characterising errors has two practical advantages: 1) for the engineer, quantifying an error allows for a clear signal to improve hardware and calibration routines, and 2) for the theorist, understanding physically motivated error channels allows for the development of error correction and mitigation techniques that are robust to real world noise. Both users are concerned ultimately with the same question: How faithfully can we execute arbitrary circuits on our physical devices? Extracting metrics to this end is a wide research area in quantum information, with techniques of varying scalability and implicit assumptions on underlying noise models [13]. On one end of the spectrum sits full tomography [14–17] which reconstructs quantum channels, here represented as completely positive, trace preserving (CPTP) maps on the space of density matrices,

$$\mathcal{E}(\rho) : \mathbb{C}^{d \times d} \rightarrow \mathbb{C}^{d \times d}, \quad (1.1)$$

where the dimension of the Hilbert space is $d = 2^n$ for n qubits. The CPTP map is rich in information, capturing all Markovian (those which do not depend on history) error channels, but has poor scaling with system size, generally requiring $O(d^4)$ parameters to describe, and $O(d^4)$ measurements to reconstruct. This scaling fundamentally limits tomography to small system sizes. The goal then shifts to how to extract useful information in predicting the behaviour of a system executing a circuit, without full channel reconstruction. Now, two strategies emerge: 1) we make assumptions on the underlying noise model, or 2) we actively simplify the structure of the error channel through averaging. The first strategy leads to techniques such as noise learning [18, 19] and the latter leads to random circuit methods, such as randomised benchmarking (RB) [10, 20, 21]. Both approached dramatically

reduce the number of parameters required to describe the system, and the number of measurements required to extract them.

The assumptions on the noise model are cleanly motivated by considering the modularity of quantum circuits, and how errors may violate this modularity. Following the description of imperfect computation and crosstalk errors given in references [22], and [5], respectively, we define the following models of QIPs: *Ideal QIPs*, *Markovian QIPs* (containing the subset of *Crosstalk-Free Markovian QIPs*), and *Non-Markovian QIPs*.

The most desirable, yet technically unfeasible, is the *Ideal QIP* — described by a closed quantum system where the qubits comprise a quantum state represented by a single vector in Hilbert space. The state is isolated from the environment and may be decomposed into a collection of subsystems that remain uncoupled unless explicit multi-qubit operations are applied. Operations are all unitary evolutions, conveniently described by single and two-qubit unitaries, which we term *gates*. Errors may exist, but are limited to reversible unitary interactions.

Markovian QIPs include external and control field noise due to an environment with no history, i.e. a source of Markovian noise. We must depart from the quantum state vector and unitary evolution description, and instead use density matrices and CPTP maps to describe the state and dynamics of the QIP. We highlight a subset of these models, termed *Crosstalk-Free Markovian QIP*, where the dynamics of distinct subsystems are governed by locally acting CPTP maps. To satisfy this condition, operations are said to be *local* and *independent* [5].

- *Local* is the assumption that only explicitly applied multi-body operations lead to correlations between qubits that participate in the ideal multi-body operation. For a bipartite system, a local Markovian QIP executing a circuit with no explicit coupling between subsystems, A , and B , can be described by the product channel CPTP map,

$$\mathcal{E}_{AB}(\rho_A \otimes \rho_B) = \mathcal{E}_A(\rho_A) \otimes \mathcal{E}_B(\rho_B), \quad (1.2)$$

where $\mathcal{E}_A$ and $\mathcal{E}_B$ are CPTP maps acting locally on the A and B subsystems respectively.

- *Independent* is the assumption that the action of an applied gate on a subset of qubits is independent of any other gates applied in parallel on the other qubits. This implies that the CPTP map of gate G , acting on subsystem A in layer i , given by $\mathcal{E}_G^{(i)}(\rho_A)$, is identical to the action of the same gate acting on subsystem A in layer j ,

$$\mathcal{E}_G^{(i)}(\rho_A) = \mathcal{E}_G^{(j)}(\rho_A), \quad (1.3)$$

where parallel applied gates on the other subsystems may be different between layers.

With these two criteria satisfied, we may retain the composibility of gates, conveniently allowing us good prediction of the behaviour of a many-qubit deep circuit, using only characterisations of each gate in a universal gate set applied to small subsystems. If these criteria are not met, to accurately predict the behaviour of a deep circuit, each unique layer must be characterised, which is intractable.

The last class of models, included for completeness, are *Non-Markovian QIP*, where we relax the assumption that the effect of the environment has no history dependence. In this case, any instantaneous state may still be able to be described by a density matrix, however the dynamics can no longer be described by the application of a sequence of CPTP maps. To accurately characterise the system, full process tomography would need to be implemented for each possible circuit.

Practically, to ensure we can reasonably characterise our physical systems, and predict their performance on novel circuits, we must design and appropriately isolate hardware to be well approximated by the *Crosstalk-Free Markovian* models. The next section provides some background on existing techniques to verify crosstalk-free performance, and provide some quantification when the assumption is violated.

1.2 Existing approaches to correlated-noise characterisation

Existing approaches to correlated-noise characterisation span a trade-off between expressiveness and scalability. The most expressive, and arguably least scalable, are tomographic techniques, such as gate set tomography [15] or the crosstalk sensitive variation [23]. Here, the implicit assumption that the noise environment is Markovian is sufficient, and no other assumptions of underlying noise models are required. These techniques are valuable in producing detailed CPTP maps that inform not only the presence of crosstalk-free violating noise, but also offer evidence on the underlying noise process. Demonstrations are however limited to small system sizes of two-qubits in [23].

Noise learning protocols are substantially more scalable by imposing a restricted set of noise models, and fitting experimental data to these models. Correlated-noise specific variations of this approach [19, 24] apply repeated random circuits that act to simplify the action of noise channels to Pauli type errors, and then through measurements and statistical inference, likely Pauli error models are found, and crosstalk-free violating errors may be quantified. Noise learning can also be adaptable by first fitting simple models, for example when twirling over random circuits yields structured error channels, and progressively increasing the complexity of the inferred model when arbitrary circuits are run on the same device [25]. This nested-model approach provides a framework for robust characterisation of the system, however, demonstrations are still currently limited to small systems.

Similar to enforcing Pauli-type errors, randomised benchmarking techniques average the action of noise channels to impose structure to general error channels. This is achieved by twirling the action of the channel over deep circuits comprised of a chosen gate group. Clifford twirling techniques, notably the standard RB protocol [10–12], have been adapted for detecting crosstalk induced correlations by simultaneous RB [26], and correlated RB [27] protocols. Correlated RB uses independent local twirling of multiple system copies to extract multi-qubit survival decay rates with Clifford sequence length. The weight and locality of correlated

errors may then be inferred through a distance of the measured channel from a product (uncorrelated) channel. Other, non-Clifford twirled RB protocols, such as cycle-benchmarking [21] and EPLG [28] apply parallel gate sequences on multiple subsystems to extract some error applied per layer, that can then be used to infer error contributions due to correlated-noise channels. These methods have been shown to scale to 10s of qubits, and provide system level diagnostics of device performance.

We note two further streams of current research, first the application of RB in determining the unitarity of noise [29], and the extension of this work to quantifying bipartite unitary noise [30], which can induce correlations across the qubit register. Second, the use of custom circuits designed to probe crosstalk-free violations [5, 31, 32]. **Either expand section, to include why these citations are relevant, or compress section to only the highly relevant papers. Dear reader, what do you think?**

The present protocol we have developed, and demonstrated experimentally, sits firmly in the RB methodology. It however differs from simultaneous and correlated RB by providing not only a quantification of crosstalk-free violating noise, but also providing evidence for the action of the error channel on a multi-qubit system. This extra information is inferred through direct observation of population decay and coupling between the subspaces enforced by twirling over the diagonally applied single-qubit Clifford group. The inspiration of this work comes from correlation spectroscopy [33, 34] and subspace-leakage techniques [35, 36]. Correlation spectroscopy demonstrates that common-mode noise can be revealed through its action on collective and differential degrees of freedom, while subspace-leakage approaches provide a framework for defining error models not by full CPTP maps, but rather by their action on a discrete set of physically relevant subspaces. In such settings, symmetry is not merely a simplifying assumption providing structure to arbitrary channels, but rather it is the very mechanism that separates crosstalk-free violating errors from local noise. The next section details the subspaces enforced by diagonally

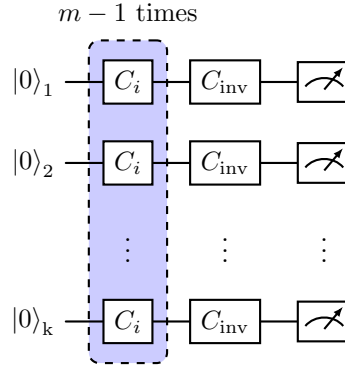


Figure 1.1: Circuit diagram for k -copy randomised benchmarking. A sequence of $m - 1$ random Clifford gates are applied in parallel to k qubits, with a final inverting Clifford applied to ideally recover the initial $|0\rangle$ state. All qubits are measured, and correlations between qubits that successfully return to the initial state are recorded.

applied single-qubit Clifford twirling, and provides the protocol for k -copy RB.

1.3 The k -copy RB scheme

This section starts with a circuit description of the k -copy RB protocol. We then develop the representation-theoretic framework required to analyze the decay curves arising from the correlated measurements. We finish this section by deriving the correlated populations observed experimentally when the system is subject to common random-unitary single-qubit, local depolarising, and entangling type noise.

1.3.1 Protocol

The k -copy RB protocol is implemented through standard single-qubit Clifford RB sequence (see section ??) applied simultaneously to multiple qubits, with the same randomly sampled Clifford applied to each qubit at every step, as shown in figure 1.1. A single sequence is composed of $m - 1$ parallel gates sampled randomly from the Clifford group. As in standard RB, the final applied layer is an inverting Clifford, which, due to the Clifford group being closed, may be selected to take the final state back to the initial computational eigenstate. The full sequence of random

Clifford gates is then equivalent to the identity operation with some unknown noise overlaid¹. Practically, each gate layer can be executed using a global field, or through parallel local control lines addressing each qubit. A final $\langle Z \rangle$ measurement is made in parallel on all qubits to probe the successful execution of the Clifford sequence. Qubits measured in the target state are labelled $\mathcal{S}$, and qubits measured in the opposite state are labelled $\mathcal{F}$, for success and failure respectively. To build statistics of success and failure rates, r random circuits are trialled for each length m , and each circuit is repeated s times to reduce shot noise. To retain information on multi-qubit correlations, for every experimental shot, data from each qubit pair (i, j) is assigned to one of four outcomes, $\mathcal{P}_{SS}^{(i,j)}$, $\mathcal{P}_{SF}^{(i,j)}$, $\mathcal{P}_{FS}^{(i,j)}$, or $\mathcal{P}_{FF}^{(i,j)}$, where S and F denote success and failure. As with other RB protocols, the success probabilities versus sequence length are then fit to explicitly defined decay curves inferred by the action of the twirled error models.

By limiting the analysis to pairwise comparisons, the *k-copy* protocol is both efficient in the number of trials required on the device, and in the postprocessing of the recovered data. For a register of k qubits, the same shot record contains outcomes for all $k(k-1)/2$ pairs. The two-body error models are derived in the following sections for prevalent noise mechanisms in both ion trapping and superconducting information processors.

1.3.2 Representations of $Cl_1^{\otimes 2}$

Here, we discuss the representations of the single-qubit Clifford group acting diagonally on two copies, $Cl_1^{\otimes 2}$, first presented in reference [37]. The irreducible representations give a natural (Schur) basis to consider how possible error channels are simplified by *k-copy* Clifford twirling.

We consider a group $\mathcal{G}$ and a representation $r : \mathcal{G} \rightarrow GL(\mathcal{L})$, where GL is the general linear group (consisting of the set of invertible matrices and the matrix multiplication operation), and $\mathcal{L}$ denotes the Liouville (vectorised operator) space

¹Note: To further symmetrise state preparation and measurement errors, we may allow the final target state to be randomised between $|0\rangle$ and $|1\rangle$ for each sequence [11].

of two qubits. Twirling over the elements g in $\mathcal{G}$ enforces symmetry on a general error channel, Λ , given in Liouville form,

$$\tilde{\Lambda} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} r(g) \Lambda r(g)^\dagger, \quad (1.4)$$

where $|\mathcal{G}|$ is the number of elements comprising the finite group $\mathcal{G}$. The twirling enforces commutation with the group action,

$$r(g) \tilde{\Lambda} r(g)^\dagger = \tilde{\Lambda} \quad \forall g \in \mathcal{G}. \quad (1.5)$$

The representation of the space, $\mathcal{L}$ decomposes to irreps λ as,

$$\mathcal{L} = \bigoplus_{\lambda} (V_{\lambda} \otimes \mathbb{C}^{m_{\lambda}}), \quad (1.6)$$

where V_{λ} is the irreducible representation space, and $\mathbb{C}^{m_{\lambda}}$ is a complex valued multiplicity space with dimension of the multiplicity m_{λ} . From Schur's lemma, equation 1.5 implies the twirled error channel has the form,

$$\tilde{\Lambda} = \bigoplus_{\lambda} (I_{\lambda} \otimes \mathcal{M}_{\lambda}), \quad (1.7)$$

where $\mathcal{M}_{\lambda}$ is the multiplicity matrix of dimensions $m_{\lambda} \times m_{\lambda}$.

The 2-copy single-qubit Clifford group, analysed in detail in references [37, 38], decomposes the general 16-dimensional two qubit space into 4 irreducible representations, irreps, with dimensions $\{1, 2, 3, 3'\}$, (the notation $3'$ is used to contrast the other 3-dimensional irrep), and multiplicities $\{2, 1, 3, 1\}$. The subrepresentations are summarised in table 1.1. **Could move character table and conjugacy classes to appendix?** The irreps are uniquely determined by their character on the conjugacy classes of the Clifford group. Two group elements $g, h \in G$ are said to be in the same conjugacy class if they are related by conjugation,

$$g = khk^{-1}, \quad (1.8)$$

where $k \in G$. With this definition, the 24 single qubit Clifford elements are divided into five permutation classes, shown in table 1.2, with a representative name and

Sub-representation	irrep	Basis operators
Identity	1	II
Left qubit	3	XI, YI, ZI
Right qubit	3	IX, IY, IZ
Trace	1	$(XX + YY + ZZ)/\sqrt{3}$
Diagonal traceless	2	$(XX + YY - 2ZZ)/\sqrt{6},$ $(XX - YY)/\sqrt{2}$
Symmetric off-diagonal	3'	$(XY + YX)/\sqrt{2},$ $(XZ + ZX)/\sqrt{2},$ $(YZ + ZY)/\sqrt{2}$
Antisymmetric	3	$(XY - YX)/\sqrt{2},$ $(XZ - ZX)/\sqrt{2},$ $(YZ - ZY)/\sqrt{2}$

Table 1.1: Sub-representations of the $Cl_1^{\otimes 2}$ group. The irrep, dimension of each sub-representation, and a set of operators in the Pauli basis which span the subspace dimension is given. The irrep basis vectors are from equation 56 in reference [38].

Class	Representation	Member
Identity	e	I
Swap	(12)	H
Two-swap	(12)(34)	X
Three-cycle	(123)	HS
Four-cycle	(1234)	S

Table 1.2: Conjugacy classes for the single qubit Clifford group.

irrep	e	(12)	(12)(34)	(123)	(1234)
Identity/Trace	1	1	1	1	1
Left/Right/Antisymmetric	3	-1	-1	0	1
Diagonal traceless	2	0	2	-1	0
Symmetric off-diagonal	3	1	-1	0	-1

Table 1.3: Character table for the Clifford group given the irreps in table 1.1, and with conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle. The “sign” irrep of the single-qubit Clifford group has zero multiplicity in the two-copy case, and so is excluded here.

example member following each class. These classes make obvious the known isomorphism between Cl_1 and the symmetric group S_4 (modulo phase). Character tables [39], are useful tools in representation theory, to uniquely describe irreps by the traces of the representative matrices for group elements of each conjugacy class. The table is presented as a 2D matrix with rows corresponding to group representations and columns to group conjugacy classes. The character, ξ , is defined by,

$$\xi_\lambda(g) = \text{Tr}[r_\lambda(g)], \quad (1.9)$$

where r_λ is the irreducible component of the representation of G acting on some vector space,

$$r_\lambda(g)_{i,j} = \text{Tr}[B_i^\dagger g B_j g^\dagger], \quad (1.10)$$

where B_i are the orthonormal basis vectors for irrep λ . Taking the two-qubit sub-representations reproduced in table 1.1, the character table calculated is shown in table 1.3. We can see confirmed that the left, right, and antisymmetric sub-representations form a multiplicity of the $3D$ -irrep labelled 3, while the trace sub-representation is a multiplicity of the $1D$ trivial irrep.

Given the structure of a general twirled map is enforced to be block diagonal in the irrep basis, given by equation 1.7, we can do a simple count to find generally how many parameters are required to fully describe the CPTP map. Accounting for the multiplicities, and the constraint that the map is trace preserving, we find that at most $2^2 + 3^2 + 1 + 1 - 2 = 13$ independent scalars are needed to describe $\tilde{\Lambda}$. The diagonal terms, here labelled $\vec{\alpha} = (\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as})$, describe a decay rate contribution from each sub-representation, whilst off-diagonal terms, $\vec{m} = (m_{id \rightarrow tr}, m_{L \rightarrow R}, m_{R \rightarrow L}, m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R})$ describe coupling between the multiplicity spaces.

In practice, different physical error mechanisms populate only a subset of the coefficients α and m , leading to simplified dynamics with characteristic signatures of the underlying process. Table 1.4 provides possible physical mechanisms capable of coupling isotypic representation subspaces. We devote some time in the next

Coupling	irrep	Mechanism
$m_{id \rightarrow tr}$	1	Decays (non-unital)
$m_{L \rightarrow R},$ $m_{R \rightarrow L}$	3	Qubit Swap, Correlated decays (non-unital)
$m_{L \rightarrow as},$ $m_{R \rightarrow as},$ $m_{as \rightarrow L},$ $m_{as \rightarrow R}$	3	Entanglement

Table 1.4: Possible mechanisms that result in the coupling of sub-representation sectors of the $Cl_1^{\otimes 2}$ twirled error channel. **Is this too strong a claim?**

two sections to describing noise due to fluctuations of a common field, and to local depolarising type errors. In appendix A.7 we discuss a ZZ -type noise model, relevant to the superconducting qubit platform.

1.3.3 Common random unitary noise

It can be seen that the error channels that lead to couplings neatly discern the physical origin of the noise channel. For example, in transmon superconducting qubit platforms, always on ZZ -couplings are a dominant crosstalk error source [40–42], and so the couplings between single qubit subspaces and the antisymmetric subspace can be quantified; in the neutral atom platform, correlated non-unital errors may be observed if the physical qubit separation is of similar order to the qubit transition wavelength [43]; however, in ion trap platforms, environmental noise is often dominated by magnetic and electric field fluctuations, which may act globally on all qubits if they are stored in the same trapping potential. These field fluctuations induce both common and independent single-qubit rotations, but do not couple irrep subspaces together (i.e $m = 0$ for $m \in \vec{m}$). Thus, the general error map of this type is now parametrised by at most the 6 scalars, $\vec{\alpha}$, and results in Liouville form diagonal in the Schur basis, $\mathcal{S}$, (ordered by table 1.1),

$$\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}) = \text{diag}(I_1, \alpha_L I_3, \alpha_R I_3, \alpha_{tr} I_1, \alpha_{dt} I_2, \alpha_{so} I_3, \alpha_{as} I_3), \quad (1.11)$$

where I_d is the d -dimensional identity.

We term noise due to common small-angle single-qubit rotations acting on k -qubits

common noise. For two qubits the general form is given by the separable channel,

$$\mathcal{E}_{\mathcal{A}}(\rho) = \sum_{a \in \mathcal{A}} p_a (U_a \otimes U_a) \rho (U_a \otimes U_a)^\dagger, \quad (1.12)$$

where $\mathcal{A}$ specifies the ensemble of unitaries applied with probabilities p_a , with $\sum_{a \in \mathcal{A}} p_a = 1$, and $U_a(\theta_a, \hat{n}_a) = e^{-i(\theta_a/2)\hat{n}_a \cdot \hat{\sigma}}$, with rotation angle, θ_a , axis, $\hat{n}_a$, and Pauli vector, $\hat{\sigma}$. A subset of error channels following this description cannot be described by a product channel map on a bipartite system, and so violate the *local* condition given in equation 1.2. These errors therefore cannot be modelled by a crosstalk-free Markovian QIP although originating from separable channels. This is an important distinction: non-product channels are not limited to direct entangling noise.

To find the twirled error channel of the common unitary channel, $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_{\mathcal{A}})$, we first consider the effect of twirling on a single common unitary channel, $\mathcal{E}_a(\rho) = (U_a \otimes U_a) \rho (U_a \otimes U_a)$. We outline two approaches to find $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_a)$:

First, as the basis sets for the relevant sub-representations are known (see table 1.1), and as the single qubit Clifford group only contains 24 elements, we may construct the common unitary channel in the Pauli transfer matrix (PTM) representation[22, 44], and, using equation 1.4, we may explicitly calculate the twirled channel $\tilde{\Lambda}^{\mathcal{P}}(\mathcal{E}_a)$ utilising a symbolic computation program². Here the superscript $\mathcal{P}$ denotes we are working in the Pauli basis. The α parameters are then simply read from the diagonal elements after transforming the PTM representation to the known basis operators for the sub-representations, i.e. transforming to $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_a)$.

The second approach utilises the symmetries of the irrep spaces, and the separability of the error channel to construct the PTM projectors onto the irrep spaces. This method is outlined in appendix A.6, and provides further insight into the symmetries each irrep possesses, and the naming convention for the sub-representations.

²Such as *Mathematica* by *Wolfram Research, Inc.*, or the *Python* package *SymPy*.

Both approaches ultimately yield the alpha parameters,

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (1.13)$$

$$\alpha_{tr} = 1, \quad (1.14)$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2\kappa), \quad (1.15)$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2\kappa), \quad (1.16)$$

where $c = \cos(\theta_a)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Taking these scalar products to second order in error angle θ_a , we find that the anisotropic term may be ignored for small errors,

$$\alpha_L = \alpha_R = \alpha_{as} = 1 - \theta_a^2/3, \quad (1.17)$$

$$\alpha_{tr} = 1, \quad (1.18)$$

$$\alpha_{dt} = \alpha_{so} = 1 - \theta_a^2. \quad (1.19)$$

Reintroducing the stochastic application of common unitaries, given by equation 1.12, we find,

$$\tilde{\Lambda}^S(\mathcal{E}_A) = \sum_{a \in \mathcal{A}} p_a \tilde{\Lambda}^S(\mathcal{E}_a). \quad (1.20)$$

As such, the dynamics for the twirled channel of an ensemble of common unitaries is entirely described by the error angle magnitudes, with alpha parameters,

$$\alpha_i = \sum_{a \in \mathcal{A}} p_a \alpha_a. \quad (1.21)$$

This definition leads to a convenient (and conforming to previous RB work) definition for the correlated error per Clifford given by,

$$\varepsilon_{\text{corr}} = \frac{1}{6} \sum_{a \in \mathcal{A}} p_a \theta_a^2, \quad (1.22)$$

where the correlated error per Clifford, $\varepsilon_{\text{corr}}$, quantifies the single-qubit error contribution due to *common noise*. Comparing to equation 1.11, the general

ensemble of common single-qubit rotations channel, after twirling over $Cl_1^{\otimes 2}$, is given by the diagonal matrix in the Schur basis,

$$\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_{\mathcal{A}}) = \text{diag}\left(I_1, (1 - 2\varepsilon_{corr})I_6, I_1, (1 - 6\varepsilon_{corr})I_2, (1 - 2\varepsilon_{corr})I_6\right). \quad (1.23)$$

1.3.4 Local depolarising noise

Ultimately, environmental noise will not perfectly couple symmetrically to both qubits, and so independent local single-qubit error channels must also be considered. Assuming an independent noise channel acting on each qubit, Clifford twirling leads to a fully depolarising noise channel of the form,

$$\mathcal{E}_{dp,i}(\rho) = (1 - 2\varepsilon_i)\rho + \varepsilon_i I_2 \text{Tr}(\rho), \quad (1.24)$$

where ρ is for a single qubit, and the subscript i signifies which qubit we are considering. For the “left” and “right” qubits we describe the decay rates by ε_L and ε_R , respectively. Parallel local depolarisation is given by the channel, $\mathcal{E}_{dp} = \mathcal{E}_{dp,L} \otimes \mathcal{I} + \mathcal{I} \otimes \mathcal{E}_{dp,R}$, with $\mathcal{I}$ being the identity channel on a single qubit. Comparing to equation 1.11, this channel may be expressed in PTM form as the diagonal matrix,

$$\Lambda^{\mathcal{P}}(\mathcal{E}_{dp}) = \text{diag}\left(1, (1 - 2\varepsilon_L)I_3, (1 - 2\varepsilon_R)I_3, (1 - 2\varepsilon_L)(1 - 2\varepsilon_R)I_9\right). \quad (1.25)$$

We find that $\Lambda(\mathcal{E}_{dp})$ already commutes with the representations $r(g)$ for all elements g in the single qubit Clifford group, as such the channel already obeys the symmetries enforced by twirling over $Cl_1^{\otimes 2}$ (see equation 1.5). Transforming to the Schur basis, we extract α parameters,

$$\alpha_L = 1 - 2\varepsilon_L, \quad (1.26)$$

$$\alpha_R = 1 - 2\varepsilon_R, \quad (1.27)$$

$$\alpha_{tr} = \alpha_{dt} = \alpha_{so} = \alpha_{as} = 1 - 2(\varepsilon_L + \varepsilon_R). \quad (1.28)$$

1.3.5 Combined model

Combining the depolarisation channels with the twirled common single-qubit rotation channel,

$$\Lambda_T(\varepsilon_L, \varepsilon_R, \varepsilon_{corr}) = \Lambda_{dp}(\varepsilon_L, \varepsilon_R) \tilde{\Lambda}_{corr}(\varepsilon_{corr}), \quad (1.29)$$

we find scalar parameters,

$$\alpha_L = 1 - 2(\varepsilon_L + \varepsilon_{corr}), \quad (1.30)$$

$$\alpha_R = 1 - 2(\varepsilon_R + \varepsilon_{corr}), \quad (1.31)$$

$$\alpha_{tr} = 1 - 2(\varepsilon_L + \varepsilon_R), \quad (1.32)$$

$$\alpha_{dt} = \alpha_{so} = 1 - 2(\varepsilon_L + \varepsilon_R + 3\varepsilon_{corr}), \quad (1.33)$$

$$\alpha_{as} = 1 - 2(\varepsilon_L + \varepsilon_R + \varepsilon_{corr}). \quad (1.34)$$

To reproduce the decay dynamics observed in the RB sequence, we must consider an initial prepared state, and the final measurement basis. For the RB sequence given in figure 1.1, the initial state is prepared as $|00\rangle$, and the final correlated populations are measured in the computational eigenstates, $\{|00\rangle \leftrightarrow SS, |01\rangle \leftrightarrow SF, |10\rangle \leftrightarrow FS, |11\rangle \leftrightarrow FF\}$, where S and F denote “success” and “failure” of recovering the target state outcome for each qubit. The correlated populations may be given analytically using the derived Liouville operators, and by the vectorized initial and final states,

$$\mathcal{P}_{LR} = \langle\langle L, R | \Lambda_T^m | 00 \rangle\rangle, \quad (1.35)$$

where L, R denote the successful recovery of the target final state for the “left” and “right” qubit respectively, and m is the number of Clifford gates in the applied sequence. Using this equation, and the explicit transfer matrices, the

final populations decays are found,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (1.36)$$

$$\mathcal{P}_{SF} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \quad (1.37)$$

$$\mathcal{P}_{SF} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \quad (1.38)$$

$$\mathcal{P}_{FF} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m. \quad (1.39)$$

Considering the final state of the single qubit subspace, as probed by standard RB, we find the expected single exponential decay,

$$\mathcal{P}_S^{(L)} = \mathcal{P}_{SS} + \mathcal{P}_{SF} = \frac{1}{2} (1 + (1 - 2(\varepsilon_L + \varepsilon_{corr}))^m), \quad (1.40)$$

where the superscript (L) refers to considering the left qubit only. Comparing to standard single qubit RB parameterisation,

$$\mathcal{P}_S = \frac{1}{2} (1 + (1 - 2\varepsilon_c)^m), \quad (1.41)$$

where ε_c is the “error-per-Clifford”, we find the observed decay of each qubit is now separable into a local, and correlated error component,

$$\varepsilon_c^{(L)} = \varepsilon_L + \varepsilon_{corr}. \quad (1.42)$$

Asymptotes

Here we consider the asymptotic structure of the correlated decay curves given in 1.36. Under dominant local depolarisation, the two-qubit state will approach the maximally mixed state, indicated by the asymptote $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$, as seen in figure 1.2 (a). Under dominant *common noise*, with decay profile seen in (b), an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/3$ is approached due to the invariance of the symmetric trace sector, α_{tr} . Panel (c) shows the case where both local depolarising and *common noise* is present. When $\varepsilon_{corr} > \varepsilon_L = \varepsilon_R$ a clear intermediate asymptote is seen before a final decay to the fully depolarised state. This intermediate asymptote is the central experimental signature of correlated noise exploited throughout this chapter, and is explored experimentally in section 1.6.2 below.

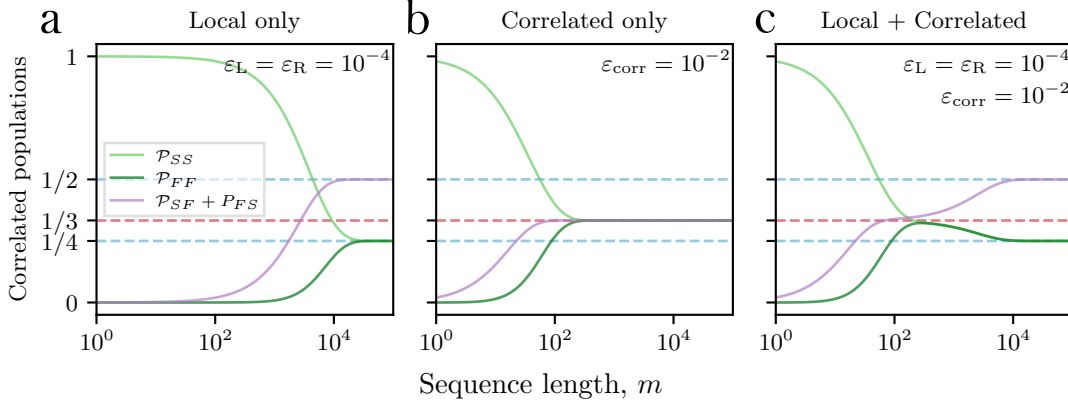


Figure 1.2: The expected multi-exponential decay profiles for two-qubit correlated success and failures recovered during *k*-copy RB. Panels (a), (b), and (c) display the structured decays for local depolarising only, *common noise* only, and the combined case, respectively. Dashed lines indicate the expected asymptotes for depolarising (blue) and for *common noise* (red).

The mixed state reached after the application of *common noise* to the initial state $|00\rangle$ can be understood by directly inspecting the space of possible pure two qubit states reached through common only rotations. The general pure state after common rotations is the separable identical two-qubit states,

$$|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle, \quad (1.43)$$

with $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$ for a single qubit general state. Depolarising into the symmetric separable two-qubit states is equivalent to averaging over the full Bloch sphere,

$$\begin{aligned} \rho_{\text{sym}} &= \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} (\sin(\theta) |\Psi\rangle \langle \Psi|) d\phi d\theta \\ &= \frac{1}{3} |00\rangle \langle 00| + \frac{1}{3} |11\rangle \langle 11| + \frac{1}{3} |\Psi^+\rangle \langle \Psi^+|, \end{aligned} \quad (1.44)$$

where the third term is the Bell state $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. The presence of this Bell-state component reflects residual coherence induced between the two qubits by the shared rotation error, and motivates the alternative strategy of a coherence-sensitive probe developed in section 1.4.

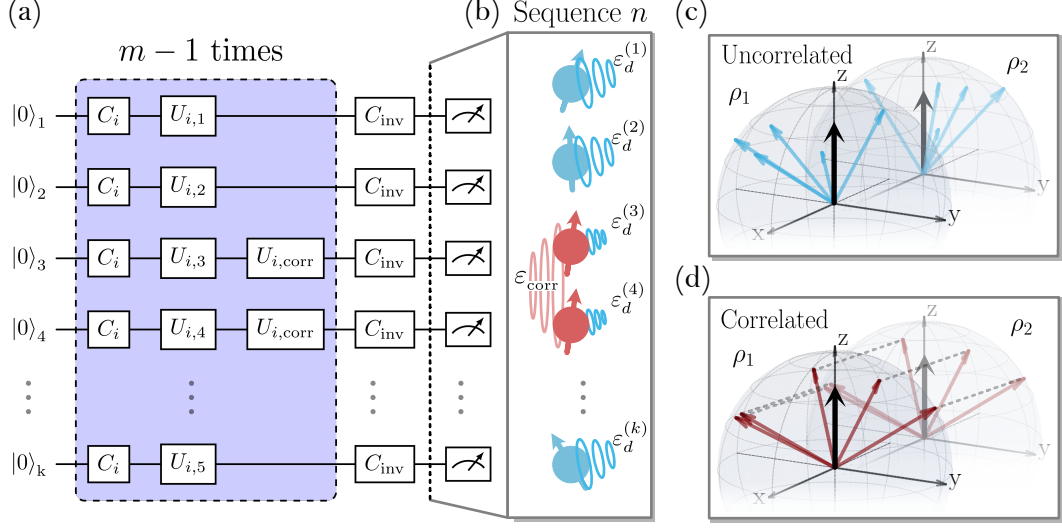


Figure 1.3: k -copy RB applies identical single-qubit Clifford sequences in parallel to all qubits. (a) shows the circuit diagram for a single RB random sequence with random unitary errors included explicitly. (b-d) illustrate the distinction between uncorrelated and *common noise* (a type of correlated noise) in the final-state trajectories.

Schematic explanation

The distinction between correlated and uncorrelated noise is illustrated schematically in figure 1.3. We consider a particularly simple unitary noise mechanism to highlight the qualitative difference of information gained between standard and k -copy RB. After each Clifford gate we consider an independent unitary rotation errors on each qubit, and a common unitary rotation error on qubits $i = 3$ and $i = 4$. Each unitary rotation axis is randomly drawn (isotropically from $SO(3)$), and the angle of the rotation is small. The error channels are explicitly included in the circuit diagram in (a). The state of the qubits after the twirling procedure, and before measurement, are shown in (b). We see that all qubits are no longer in the eigenstate $|0\rangle$ (up arrow) due to the unitary noise. Panels (c) and (d) display the final vectors resulting from 5 random sequences. All qubits here exhibit identical single-qubit RB error rates, which are probed by standard RB, and shown here through the identical depolarised Bloch vectors (black arrows). However, looking at pairwise correlations, we see: independent noise produces uncorrelated sequence-to-sequence deviations, whereas common noise produces shared deviations between qubits acted

on by the same random Clifford sequence. The parallel twirling during *k*-copy RB preserves exactly this symmetry information that would be averaged away under independently sampled local Clifford sequences.

1.3.6 State preparation errors

Similar to previous work on subspace leakage benchmarking [35], state preparation errors can lead to non-trivial modifications of the correlated population decay curves. First, we consider independent state preparation on two qubits. We model independent errors as probabilistic bit flips. To first order in error, on a single qubit we have,

$$\rho_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)}) |0\rangle\langle 0| + \varepsilon_{sp}^{(i)} |1\rangle\langle 1|, \quad (1.45)$$

where the superscript refers to which qubit, *L*, or *R*, the preparation error is relevant to. This error may be modelled by $\mathcal{E}_{sp}^{(i)}$, a probabilistic *X* channel applied to an arbitrary state,

$$\mathcal{E}_{sp}^{(i)}(\rho) = (1 - \varepsilon_{sp}^{(i)}) \rho + \varepsilon_{sp}^{(i)} X \rho X. \quad (1.46)$$

Transforming to the Liouville form, denoted by $\Lambda_{sp}^{(i)}$, this error is simply incorporated into the population dynamics given by equation 1.35, by acting on the initial state,

$$\mathcal{P}_{LR} = \langle\langle L, R | \Lambda_T^m \Lambda_{sp}^{(L)} \Lambda_{sp}^{(R)} | 00 \rangle\rangle. \quad (1.47)$$

In this picture, the state preparation errors scale only the overlap of the initial state with the sub-representations. The overlaps are multiplied by,

$$A_{sp}^{(i)} = (1 - 2\varepsilon_{sp}^{(i)}), \quad (1.48)$$

if the qubit is involved non-trivially in the subspace projector, i.e. for the left and right qubit sectors $A_{sp} = A_{sp}^{(i)}$. Relevant to this work, the correlated survival populations with state preparation errors included are,

$$\begin{aligned} \mathcal{P}_{SS} = & \frac{1}{4} + \frac{A_{sp}^{(L)}}{4} \alpha_L^m + \frac{A_{sp}^{(R)}}{4} \alpha_R^m \\ & + \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{6} \alpha_{dt}^m + \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{12} \alpha_{tr}^m. \end{aligned} \quad (1.49)$$

Similar forms exist for populations, $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$. Significantly, this state preparation error does not add projection on the antisymmetric or symmetric off-diagonal sectors, which would introduce another exponential component into the observed decay curves.

Correlated state preparation errors may be considered by adapting equation 1.46 to a two-qubit stochastic error channel, such as correlated spin-flips given by

$$\mathcal{E}_{sp}^{corr}(\rho) = (1 - \varepsilon_{sp}^{corr})\rho + \varepsilon_{sp}^{corr}(X \otimes X)\rho(X \otimes X). \quad (1.50)$$

The subspaces involving both qubits non-trivially are now unaffected by the state preparation error, shown explicitly by,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{A_{sp}^{corr}}{4}\alpha_L^m + \frac{A_{sp}^{corr}}{4}\alpha_R^m + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (1.51)$$

where $A_{sp}^{corr} = 1 - 2\varepsilon_{sp}^{corr}$. These correlations thus appear similar to *common noise* errors in the invariance of α_{tr} to this channel. Therefore correlated state prep errors can lead to a systematic bias in quantification of correlated error per Clifford if not accounted for in the fit by an initial state preparation error offset.

1.4 Parity contrast: an independent probe of coherence

Population decays described by equation 1.36 indicate an intermediate asymptote corresponding to the suggested state, ρ_{sym} , given by equation 1.44. Here we propose a parity-based measurement to independently verify the coherent structure seen in ρ_{sym} . Specifically, a partial state-tomography method based on [45] is used to quantify overlap of the final measured state with the Bell state, $|\Psi^+\rangle$, component in ρ_{sym} .

The circuit for probing this coherence is shown in figure 1.4 (a): differential rotations of the form $\hat{R}_{+\phi}(\pi/2) \otimes \hat{R}_{-\phi}(\pi/2)$, with rotation axis $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$, are applied prior to measurement, and the contrast of the resulting fringe (as ϕ is varied) is extracted.

Figure 1.4(b) show the expected contrast of parity measurements as a function of

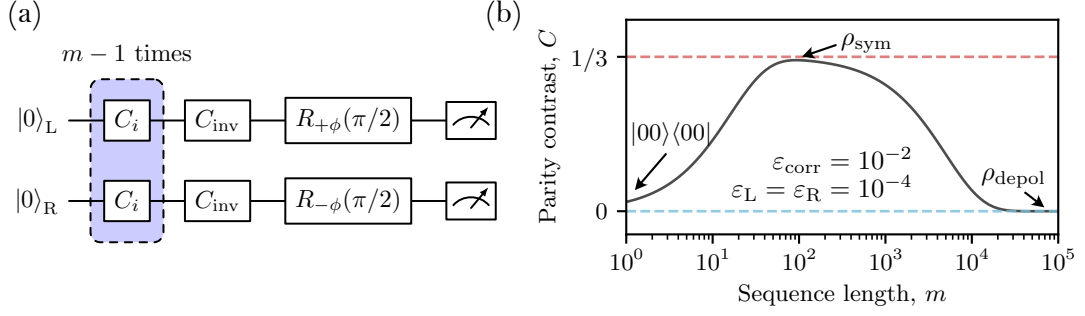


Figure 1.4: Probing coherence of the mixed state after k -copy RB. Panel (a) shows a circuit that first twirls the state over $Cl_1^{\otimes 2}$, and then applies differential $\pi/2$ -pulses with variable phase, ϕ , to recover parity oscillations. Panel (b) shows the expected contrast of the oscillations as a function of sequence length when both local depolarising and *common noise* is present. The dashed lines indicate expected asymptotes for depolarising (blue) and for *common noise* (red).

Clifford sequence length. The analytic model of the parity contrast can be simply calculated by extending equation 1.35 to include the final differential rotation, in transfer matrix form, Λ_Π ,

$$\begin{aligned}\Pi(\phi) &= \langle \langle \pi | \Lambda_\Pi(\phi) \Lambda_T^m | 00 \rangle \rangle, \\ &= C \cos(2\phi),\end{aligned}\tag{1.52}$$

where Π is the parity signal, $|\pi\rangle\rangle = |00\rangle\rangle + |11\rangle\rangle - |01\rangle\rangle - |10\rangle\rangle$, and C is the parity contrast. Evaluating $\Pi(\phi)$ for *common* and depolarising noise, we find the parity contrast follows,

$$C = \frac{1}{3}(\alpha_{tr}^m - \alpha_{dt}^m),\tag{1.53}$$

with α_{tr} and α_{dt} given by equation 1.30. This method is demonstrated experimentally in section 1.6.3 below.

Furthermore, we may now isolate some α decay parameters by combining the contrast measurements with the computational state measurements made in k -copy RB. Namely, if we define the computational parity, Π_Z , as

$$\begin{aligned}\Pi_Z &= \mathcal{P}_{SS} + \mathcal{P}_{FF} - \mathcal{P}_{SF} - \mathcal{P}_{FS} \\ &= \frac{1}{3}(2\alpha_{dt}^m + \alpha_{tr}^m),\end{aligned}\tag{1.54}$$

then in combination with C , we may isolate single exponential decays originating from the independent sectors named “symmetric trace”, and “diagonal traceless”.

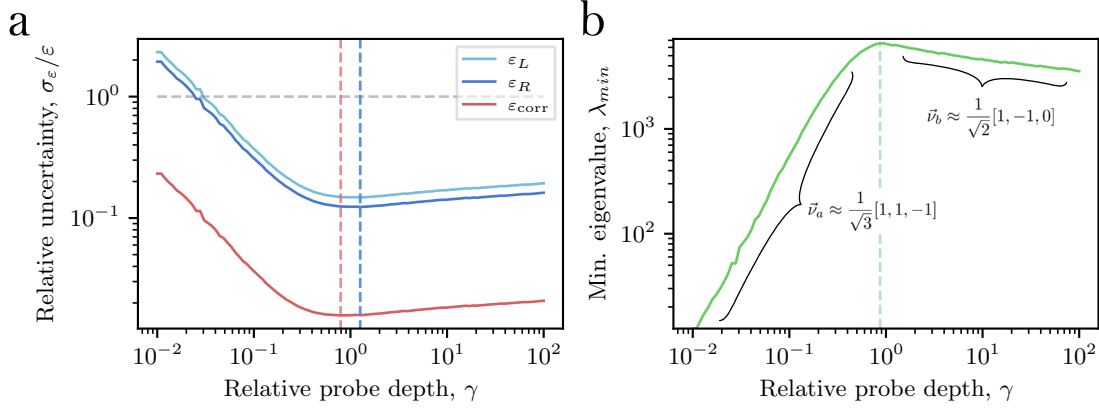


Figure 1.5: The performance of the *k*-copy fitting as a function of maximum Clifford sequence length (probe depth). Probe depth, γ , is relative to the characteristic decay length for the slowest mode, α_{tr}^m . Panel (a) uses the Cramér-Rao derived relative uncertainty for each extracted ε , while (b) shows the minimum eigenvalue of the Fisher information matrix, with associated eigenvectors highlighted.

Specifically the combinations

$$\begin{aligned}\Pi_Z - C &= \alpha_{dt}^m, \\ \Pi_Z + 2C &= \alpha_{tr}^m.\end{aligned}\tag{1.55}$$

1.5 Sampling strategy

In this section we find the Fisher information gained from a single shot of the *k*-copy RB protocol, and use this to find a strategy for setting the sequence lengths, m . We then explore the shot efficiency of extracting error per Clifford parameters for a range of underlying noise magnitudes, using both the *k*-copy protocol described in section 1.3, and using the parity measurement protocol described in section 1.4.

The Fisher information [46], $\mathcal{I}$, quantifies what we can learn from random observable, X , about a set of underlying unknown parameters, θ , that models the probability of observing X . It is particularly informative to consider the Cramér-Rao bound, which states that the inverse of $\mathcal{I}$ is a lower bound for the variance of any unbiased estimator of θ ,

$$\text{Cov}(\theta) \gtrsim \mathcal{I}(\theta)^{-1}.\tag{1.56}$$

The Fisher matrix is defined as,

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_X \left[\frac{\partial}{\partial \theta_i} \log f(X; \theta) \frac{\partial}{\partial \theta_j} \log f(X; \theta) \right], \quad (1.57)$$

where $f(X; \theta)$ is the likelihood function, and θ is a set of parameters. Here, we are interested in a multinomial probability function, with each shot resulting in a categorical outcome $X \in \{SS, SF, FS, FF\}$. The population functions given by equations 1.36, are then the probability of observing $X = x$ for a given set of parameters, $\theta = \{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$, yielding,

$$f(X = x; \theta) = P_x(\theta). \quad (1.58)$$

We now find the Fisher matrix to be,

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_X \left[\frac{1}{P_X^2} \frac{\partial P_X}{\partial \theta_i} \frac{\partial P_X}{\partial \theta_j} \right]. \quad (1.59)$$

For a discrete random variable, the expectation is,

$$\mathbb{E}[g(X)] = \sum_x P_x g(x), \quad (1.60)$$

giving our final expression for the single-shot multinomial Fisher matrix,

$$\mathcal{I}_{ij} = \sum_{x \in \{SS, SF, FS, FF\}} \frac{1}{P_x} \frac{\partial P_x}{\partial \theta_i} \frac{\partial P_x}{\partial \theta_j}. \quad (1.61)$$

The information gained in any one shot depends on the underlying error magnitudes, as well as the chosen sequence length, m . As the correlated profiles, P_x , appear as multi-exponentials decays dependent on common parameters, θ , we must choose a range of probe sequence lengths that yield large information over all the characteristic decays. As the characteristic decays are unknown, a good heuristic strategy for fitting an unknown exponential is logarithmic sampling. However, the total extent of sequence lengths to test is non-obvious. It would be experimentally simple if the initial gradient of the correlated profiles successfully estimated the parameter θ , however, expanding the populations around $m = 0$, for small errors, we find that both purely uncorrelated and purely *common* noise decays are indistinguishable,

$$\mathcal{P}_{SS,LR}|_{m=0} \approx \mathcal{P}_{SS,\text{corr}}|_{m=0} \approx 1 - 2\varepsilon, \quad (1.62)$$

where we have assumed identical error magnitudes, $\varepsilon_L = \varepsilon_R = \varepsilon_{\text{corr}} = \varepsilon$. This suggests that we must collect data near the asymptotic structure of the decays to distinguish uncorrelated and *common* noise sources. To verify this, we set experimentally reasonable parameters for an underlying *common noise* dominated decay, and vary the maximum sequence length, m_{max} , while observing the expected uncertainty given by the Cramér-Rao bound of the Fisher information. We normalise by the magnitude of the parameter estimated to give the relative uncertainty, $\frac{\sigma_\varepsilon}{\varepsilon}$. We also find the minimum eigenvalue of the Fisher matrix, λ_{min} , which corresponds to how accurately the least constrained vector in parameter space can be measured. Figure 1.5 uses $N = 5000$ samples per sequence length (e.g. $s = 50$ shots, and $r = 100$ random sequences), and 10 total sequence lengths sampled logarithmically from $m_{\text{min}} > 2$ to $m_{\text{max}} = \gamma\tau$. The scalar γ is the relative probe depth and τ is the characteristic decay for the slow, α_{tr} mode,

$$\tau = \frac{-1}{\log(\alpha_{tr})} = \frac{-1}{\log(1 - 2(\varepsilon_L + \varepsilon_R))}. \quad (1.63)$$

We set error magnitudes of $\varepsilon_L = 1.0 \times 10^{-3}$, $\varepsilon_R = 1.2 \times 10^{-3}$, $\varepsilon_{\text{corr}} = 1.0 \times 10^{-2}$, which are similar magnitudes we observe experimentally (see section 1.6). It can be seen that the relative uncertainty for all θ are minimised when $\gamma \approx 1$. This length also corresponds to maximising λ_{min} — lowering the cost of estimating the least constrained vector of parameters, where the vector is written in the basis $[\hat{\varepsilon}_L, \hat{\varepsilon}_R, \hat{\varepsilon}_{\text{corr}}]$. It is interesting to note that the least constrained parameter vector, when $\gamma \ll 1$, roughly corresponds to $\vec{\nu} \approx \frac{1}{\sqrt{3}}[1, 1, -1]$, confirming our suspicion that *common* and depolarising noise is indistinguishable at small m . In the case that $\gamma \gg 1$, we find $\vec{\nu} \approx \frac{1}{\sqrt{2}}[1, -1, 0]$, corresponding to the depolarisation rates on the left and right qubit are tending to be indistinguishable, however we note that even over a couple of orders of magnitude, the distinguishability of ε_L and ε_R does not dramatically drop. Here, we draw the conclusion that a good strategy for efficient information per shot, is to set m_{max} to the characteristic decay of the slowest mode.

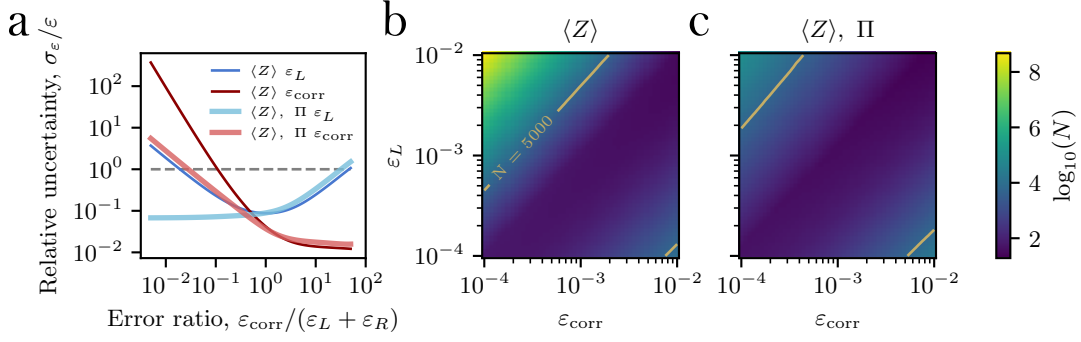


Figure 1.6: (a) The Cramér-Rao derived relative uncertainty for a range of correlated and uncorrelated error ratios. The thin lines correspond to expected uncertainties using $\langle Z \rangle$ measurements from *k-copy* RB, while thick lines use both $\langle Z \rangle$ and parity measurements, as described in section 1.4. (b-c) The required number of shots, N , per sequence length, m , for unity fractional uncertainty, for a range of correlated and uncorrelated errors. In (b), only $\langle Z \rangle$ measurements are used, while in (c) both $\langle Z \rangle$ and parity measurements are used. A contour is shown for 5000 samples, which is typically used in our experiments.

We now explore the ability for *k-copy* RB to fit θ for a range of underlying error magnitudes. Figure 1.6(a) shows the relative uncertainty as a function of the error ratio, defined here as $\varepsilon_{\text{corr}}/(\varepsilon_L + \varepsilon_R)$. We fix $\varepsilon_L = \varepsilon_R = 1 \times 10^{-3}$, and again set $N = 5000$ samples per sequence length m , with $m_{\text{max}} = \tau$, and vary $\varepsilon_{\text{corr}}$ to vary the error ratio. The thin blue and red line show the expected relative uncertainty for ε_L and $\varepsilon_{\text{corr}}$ respectively. We can see that for the case that local noise dominates, $\varepsilon_L \gg \varepsilon_{\text{corr}}$, the sufficient information for accurately estimating the correlated noise is not available. This limitation was found to be experimentally important, and motivates the injection of fixed *common noise* offsets in sections 1.6.4, and 1.6.5. Another strategy to alleviate this problem is by including data from the coherence probes described in section 1.4. The thick blue and red lines correspond to measurements where $N = 2500$ samples are collected by the typical $\langle Z \rangle$ *k-copy* protocol, and $N = 2500$ samples are using the parity protocol, denoted as Π . We find that inclusion of the parity samples dramatically reduce the relative uncertainty of the fitted parameters when $\varepsilon_L \gg \varepsilon_{\text{corr}}$. Panels (b) and (c) show the required number of samples, N , for a relative uncertainty $\frac{\sigma_\varepsilon}{\varepsilon} = 1$, for a range of local and *common* noise magnitudes. We highlight the $N = 5000$ sample contour as this may be collected in reasonable time on our apparatus. Again, we see that including

measurements using the parity protocol dramatically reduces the required number of samples in the regime of $\varepsilon_L \gg \varepsilon_{\text{corr}}$ (top left corner of plots).

Here we have shown that the Fisher information gathered in an experimentally realistic *k*-copy protocol is sufficient to accurately determine the underlying error per Cliffords, $\theta = \{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$, over many orders of magnitude of local and *common* errors. We shall show experimentally in section 1.6.4 that in the presence of experimental imperfections, and the fitting protocol (Levenberg-Marquadt), whose effects are not considered in this analysis, the proposed protocol still sufficiently recovers the magnitudes of the expected error per Cliffords with known noise injection.

1.6 Experimental Results

The results presented here constitute the first project-data output from the newly commissioned *FastGates* apparatus. This work was both necessary for confirming the utility of the *k*-copy RB protocol, as well as pushing the automation of the apparatus for robust operation, useful for all future experimental projects.

1.6.1 Physical implementation

We demonstrate *k*-copy RB using up to eight trapped $^{40}\text{Ca}^+$ ions separated by an average spacing of $\sim 5 \mu\text{m}$. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ in a quantising field of $\sim 5 \text{ G}$. Single-qubit rotations are driven by 729-nm light, with details given in section ???. The two-qubit coherence and Stark shift measurements use the individual addressing system, detailed in section ??, while all other measurements use the global elliptical beam, detailed in section ??.

The single-qubit Clifford gate set is decomposed into native $\pi/2$ single-qubit rotations of the form $\hat{R}_x(\pi/2)$ and $\hat{R}_z(\pi/2)$: $\hat{\sigma}_x$ rotations are performed natively on resonance with the qubit transition, with relative phase set by modulating the 729-nm beam

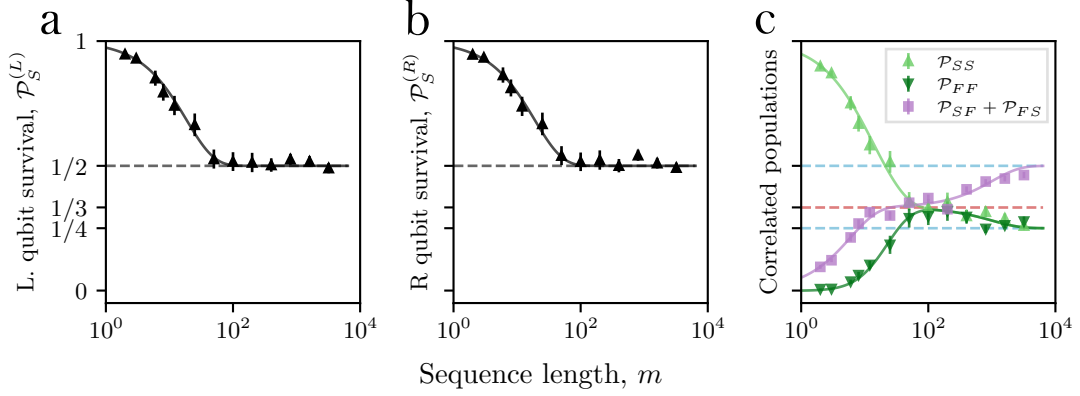


Figure 1.7: Correlated two-qubit probabilities after *k*-copy RB. Panels (a,b) show the single-qubit success probabilities, which decay with a single exponential (solid line) toward $\mathcal{P}_S|_{m \rightarrow \infty} = 1/2$. Panel (c) shows the correlated two-qubit populations, which exhibit an intermediate $\mathcal{P}_{SS} = 1/3$ asymptote before relaxing toward $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$.

via an acousto-optic modulator (AOM), while $\hat{\sigma}_z$ rotations are performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations. Two decomposition strategies are used, crucially with varying average numbers of physical $\hat{\sigma}_x$ $\pi/2$ -gates per Clifford. The first uses an average of $\kappa = 2.17$ $\pi/2$ -gates, while the second uses $\kappa = 1.167$ gates.

We begin by reproducing the multi-exponential decays seen in simulation, figure 1.2, followed by parity measurements to observe the coherence of the intermediate symmetric state described in section 1.4. We then present data with known injected errors through phase and detuning offsets to show the ability for *k*-copy RB to discern correlated and uncorrelated errors for a range of magnitudes. We finish this section with evidence for scaling the protocol to more than two qubits. We show both data for three-qubit decay profiles, and pairwise analysis of an eight-ion chain with dominant correlated errors due to global drive field inhomogeneity.

1.6.2 Z-basis measurements

The *k*-copy RB protocol, with circuit diagram shown in figure 1.1, is executed on two qubits using a global field. To ensure a visually clear intermediate asymptote is seen, we set a fixed 10 kHz frequency detuning of the 729-nm laser compared to the bare qubit. As single qubits gates are calibrated for on-resonance pulses,

this detuning will be a large *common noise* source. The Clifford sequence length is log-sampled up to $m = 3200$ total gates to capture both the intermediate asymptote due to this common unitary error, and the final decay to the fully depolarised state. The recovered multi-exponential decay from the correlated success and failure rates of the two qubits is shown in figure 1.7(c). Each solid point is inferred through $r = 50$ randomisations and $s = 50$ shots per randomisation. For each random sequence j , the population $\mathcal{P}_{SS,j}$ is estimated from the s binary measurement outcomes, $\mathcal{P}_{SS,j} = \frac{1}{s} \sum_{i=1}^s X_{ij}$, where $X_{ij} = 1$ if the i th shot is measured in state SS , and $X_{ij} = 0$ otherwise. The recovered population is then the mean over the r random sequences, $\mathcal{P}_{SS} = \frac{1}{r} \sum_{j=1}^r \mathcal{P}_{SS,j}$. Error bars show the standard error of the mean across random sequences,

$$\text{SE}(\mathcal{P}_{SS}) = \sqrt{\frac{1}{r^2} \sum_{j=1}^r (\mathcal{P}_{SS,j} - \mathcal{P}_{SS})^2}. \quad (1.64)$$

The correlated populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|00\rangle$ approaching ρ_{sym} , equation ??, with $\mathcal{P}_{SS} \rightarrow \frac{1}{3}$ asymptote highlighted by the red dashed line. The dashed blue line shows the fully depolarised $\mathcal{P}_{SS} \rightarrow \frac{1}{4}$ asymptote, seen clearly by $m \approx 1000$ gates. The solid lines are found through modelling errors by a combination of local depolarising errors and *common noise*, given by equation 1.36 generalised to include independent state-preparation and measurement error (SPAM), ε_{sp} (see equation. 1.49 for example). The model is fit using the non-linear Levenberg–Marquardt method, yielding extracted parameters $\varepsilon_{sp} = 0(5) \times 10^{-3}$, $\varepsilon_{\text{corr}} = 2.6(2) \times 10^{-2}$, $\varepsilon_L = 5(6) \times 10^{-4}$, and $\varepsilon_R = 0(6) \times 10^{-4}$, confirming that correlated errors dominate with this global field detuning.³

From the same shot data record we recover the single-qubit populations required for standard RB, as described in section ??. To match the naming of the relevant participating irreps, we label the two qubits L and R for left and right respectively.

³The large relative uncertainty of both local depolarising rates is due to the large covariance between ε_L and ε_R . The fitting may be constrained to the sum of local errors, to find $\varepsilon_L + \varepsilon_R = 5(1) \times 10^{-4}$. However, if precision measurements of this local error is of interest, either r , and s could be increased, or the local error contributions could be isolated as discussed in section ??.

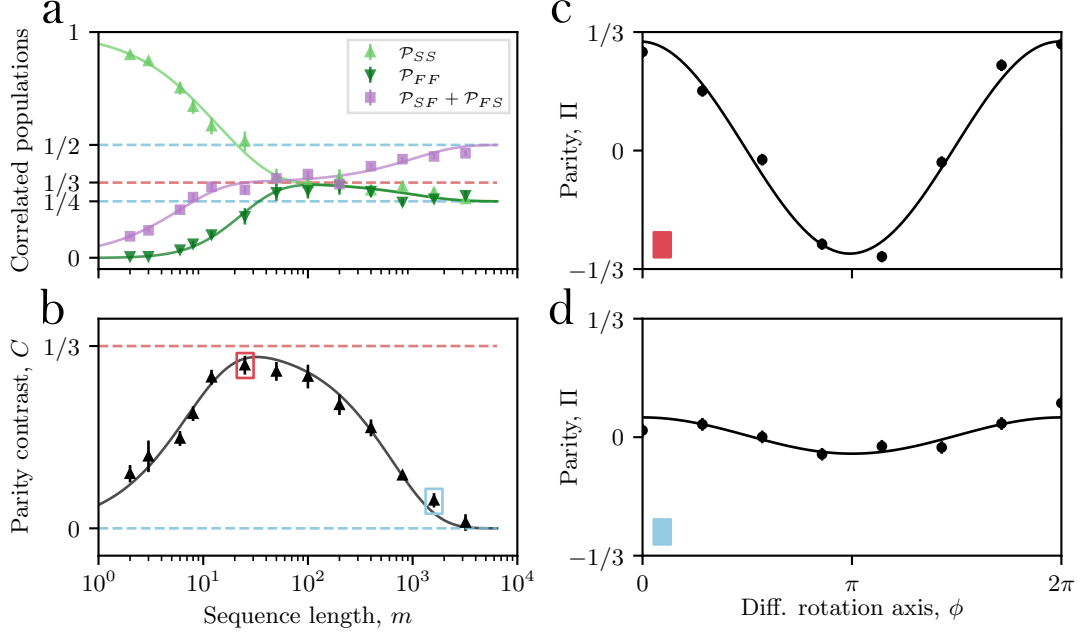


Figure 1.8: Parity contrast versus sequence length; (a) shows the standard Z measured correlated populations. (b) shows parity contrast — a signal proportional to occupation of the symmetric-trace sector. Solid lines are multi-exponential decays given by equation 1.36 and equation 1.53 for (a) and (b) respectively. Dashed lines indicate the expected asymptotes. (c) and (d) show example parity fringes for sequences of $m = 25$ and $m = 1600$ Clifford gates respectively with solid lines given by equation 1.52.

The standard RB single exponential decay curves are shown in (a) and (b), with extracted single-qubit error per Cliffords of $\varepsilon_c^{(L)} = 2.6(2) \times 10^{-2}$ and $\varepsilon_c^{(R)} = 2.6(2) \times 10^{-2}$. As expected, *k*-copy RB provides evidence for this standard error per Clifford to be decomposed to local and correlated components, given by equation 1.42.

1.6.3 Coherence measurements

A partial-tomography parity measurement, given by the circuit seen in figure 1.4(a), is used to independently confirm occupation of the symmetric sector, quantifying overlap of the final state with Bell state, $|\Psi^+\rangle$. As the final differential rotation pulse requires local addressing of the ions, the single addressing system is used. Each random sequence is repeated for varying final rotation angle, ϕ , to produce parity fringes, examples shown in (c) and (d) in figure 1.8 for $m = 25$ and $m = 1600$ Clifford gates respectively. The contrast is extracted by equation 1.53, and a plot against

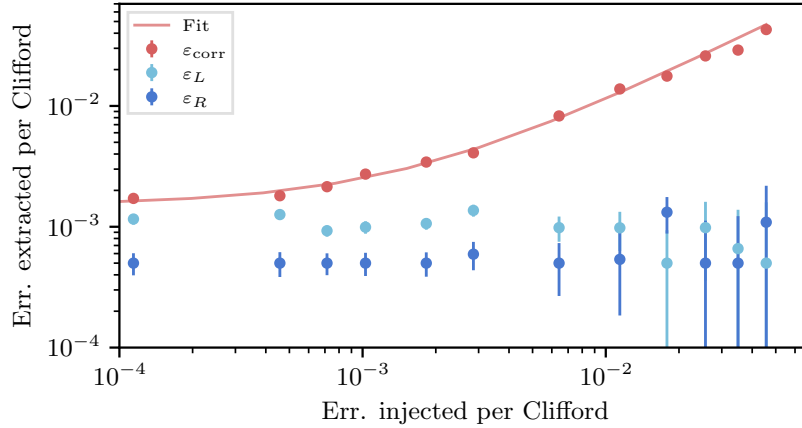


Figure 1.9: Injecting correlated noise between two qubits via common drive phase noise. Points correspond to fitted error per Clifford from full *k*-copy RB experiments. The solid red line is a linear fit between measured and injected error.

sequence length is shown in (b). The contrast asymptote $C \rightarrow 1/3$ is observed for $m \in [10, 100]$, corresponding to the residual coherence of ρ_{sym} , equation 1.44. The parity contrast data is fit (solid line) according to equation 1.53 to extract error per Cliffords of, $\varepsilon_{\text{corr}} = 2.1(2) \times 10^{-2}$ and $\varepsilon_L = \varepsilon_R = 3.9(3) \times 10^{-4}$.

1.6.4 Controlled noise injection

To verify the ability for *k*-copy RB to discern correlated and uncorrelated errors, we compare the magnitude of extracted errors to known injected errors. We limit the injected error method to those consisting of noise on common drive fields, and local depolarising type noise. Both of these noise models can be induced entirely through single qubit rotations, which are trivial to implement on our device.

Laser phase jitter

We first inject correlated errors via phase noise on a common drive field. This noise mechanism is prevalent in many platforms that use a common oscillator, such as laser, microwave, or radio-frequency (RF) fields to resonantly drive qubit transitions.

The injected error magnitude is inferred through considering the ideal and noisy unitary channel. A unitary qubit rotation is given by,

$$\hat{U} = e^{-i\frac{\theta}{2}\vec{n}\cdot\vec{\sigma}}, \quad (1.65)$$

where θ is the rotation angle, and $\vec{n} \cdot \vec{\sigma}$ is the rotation axis. We decompose the single-qubit Clifford gate set into a series of resonant $\theta = \pi/2$ qubit rotations around axes $\hat{\sigma}_x$ and $\hat{\sigma}_z$. Practically, phase noise on either the qubit frequency or drive field will introduce some deviation of this rotation axis. We shall consider the effect of zero-mean phase noise on $\hat{\sigma}_x$ rotations, leading to an erroneous rotations axis,

$$\hat{\sigma}_{\delta\phi} = \cos(\delta\phi)\hat{\sigma}_x + \sin(\delta\phi)\hat{\sigma}_y, \quad (1.66)$$

where $\delta\phi$ is the standard deviation of the noise in radians. We may find the expected error of a single Clifford gate using,

$$\varepsilon_{\delta\phi} = \kappa(1 - \mathcal{F}(V, U)), \quad (1.67)$$

where κ is the average number of $\pi/2$ rotations in a Clifford gate, and $\mathcal{F}(V, U)$ is the average gate fidelity. The fidelity of two unitaries averaged over the Haar measure is given in appendix equation A.3, reproduced here,

$$\mathcal{F}(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right). \quad (A.3)$$

Considering $\theta = \pi/2$ rotations, we find,

$$\hat{U} = e^{-i\frac{\pi}{4}\hat{\sigma}_x} = \frac{1}{\sqrt{2}} \left(\hat{I} - i\hat{\sigma}_x \right), \quad (1.68)$$

$$\hat{V} = e^{-i\frac{\pi}{4}\hat{\sigma}_{\delta\phi}} = \frac{1}{\sqrt{2}} \left(\hat{I} - i\hat{\sigma}_{\delta\phi} \right). \quad (1.69)$$

The fidelity, given that dimension $d = 2$ for a single qubit, is found to be,

$$\mathcal{F}(V, U) = \frac{1}{6} \left((1 + \cos(\delta\phi))^2 + 2 \right). \quad (1.70)$$

For small phase noise standard deviations, $\delta\phi \ll 1$, we find an expected error per Clifford of,

$$\varepsilon_{\delta\phi} = \frac{\kappa}{3}(\delta\phi)^2. \quad (1.71)$$

A known phase error may be conveniently injected through phase offsets sampled from a normal distribution with zero mean and standard deviation $= \delta\phi$ applied to the RF drive of a control AOM on the 729-nm global addressing path. The phase of the AOM RF is imprinted directly onto the 729-nm light leading to the expected drop in fidelity of the Clifford gates. In this experiment $\kappa = 2.17 \pi/2$ rotations. A full *k-copy* RB experiment is executed with logarithmic sampling up to $m = 512$ Clifford gates with $r = 50$ randomisations and $s = 50$ repeats. The resulting multi-exponential success and failure probability decay curves are fit to equation 1.36, again generalised to include state-preparation errors. The correlated and local error components are found for a range of injected phase noise standard deviations. The resulting measured errors are shown in figure 1.9. The x -axis is the expected error injected $\varepsilon_{\delta\phi}$, while the y -axis is the extracted error magnitudes. The fit, shown by the solid red line, is a direct proportionality of injected error plus a baseline correlated error rate not due to phase injection,

$$\varepsilon_{\text{corr}} = \varepsilon_{\text{corr},0} + \varepsilon_{\delta\phi}. \quad (1.72)$$

The baseline correlated error is therefore the only parameter to fit. With $\varepsilon_{\text{corr},0} = 1.71(4) \times 10^{-3}$, we find good agreement between the model and data. The local depolarising error on both qubits remains relatively constant, with the left qubit having a larger local error compared to the right qubit. We believe this asymmetry in local error is due to the global 729-nm beam not being centrally aligned to both qubits. The uncertainty in extracted error is found through Levenberg-Marquadt non-linear fit, it can be seen that the uncertainty in the depolarising errors increases substantially with the total correlated error. This may be explained by the available Fisher information gained by the chosen logarithmic sampling strategy, as explored in section 1.5. Figure 1.6 (a) supports that we expect an increase in depolarising parameter uncertainty for $\varepsilon_{\text{corr}} \gg (\varepsilon_L + \varepsilon_R)$.

Laser detuning scan

k-copy RB can isolate errors arising from shared control fields, making it useful both for device characterisation and for optimising global control parameters. A fixed detuning of the global drive, under the action of *k-copy* Clifford twirling, is well modelled by the common random unitary error channel described in section 1.3.3. We demonstrate this by scanning the 729-nm laser frequency used for single-qubit rotations away from the bare qubit resonance. As with the previous section, we consider the fidelity of the error channel compared to the ideal $\pi/2 \hat{\sigma}_x$ single qubit rotation.

A coherent detuning error is written as,

$$V = \exp \left(-i \frac{\pi}{4} \left(\hat{\sigma}_x + \frac{\delta}{\Omega} \hat{\sigma}_z \right) \right), \quad (1.73)$$

where δ is the detuning from resonance, and Ω is the Rabi frequency. Assuming $\delta/\Omega \ll 1$, the fidelity is found to be,

$$\mathcal{F}(V, U) = 1 - \frac{1}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (1.74)$$

To convert this to an error contribution to a single qubit Clifford gate, we introduce the scalar κ_δ , to account for both the number of $\pi/2$ rotation in each Clifford gate, but also the duration and delays between pulses where phase tracking between qubit and laser also drift. The error contribution is then given by

$$\varepsilon_\delta = \kappa_\delta (1 - \mathcal{F}(V, U)) = \frac{\kappa_\delta}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (1.75)$$

Each data point is obtained from a complete *k-copy* RB experiment with 11 sequence lengths up to $m = 1024$, using $r = 50$ random sequences and $s = 50$ shots per sequence, with correlated populations fit to equation 1.36 to extract $\{\varepsilon_{\text{corr}}, \varepsilon_L, \varepsilon_R\}$ at each detuning. The upper panels of figure 1.10 show example datasets for the *k-copy* RB sequences, while the lower panel shows the extracted error per Cliffords as a function of detuning. The correlated error is observed to increase quadratically with detuning, while the local depolarising errors remain approximately constant, consistent with detuning of a shared drive producing *common noise*. To

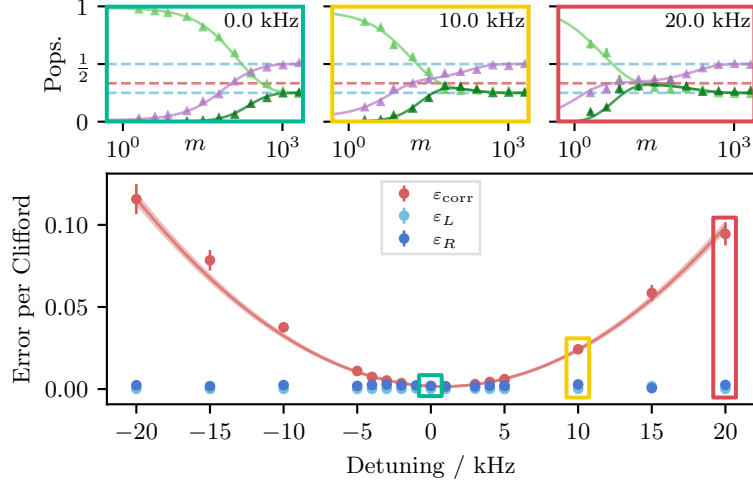


Figure 1.10: Correlated and single-qubit depolarising errors extracted as a function of shared-drive detuning. The correlated error (red points) rises quadratically with laser detuning, while the local depolarising errors (blue points) remain approximately constant. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. The red shaded region signifies the 68% confidence, generated through covariance-based parameter sampling.

fully parameterise the injected error, we include a background correlated error not originating from detuning, ε_0 , and a shift of the light-shifted qubit versus the bare qubit frequency, δ_0 . The solid red line is therefore fit to the model,

$$\varepsilon_{corr} = \varepsilon_0 + \frac{\kappa_\delta}{3} \left(\frac{\delta_L - \delta_0}{\Omega} \right)^2. \quad (1.76)$$

Here, $\Omega = 2\pi \times 100(1)$ kHz is the independently measured Rabi frequency. Fitting gives $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $\kappa_\delta = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)$ kHz.

Single addressed Stark shifts

Uncorrelated noise injection requires the introduction of noise with spatial inhomogeneity. This is implemented experimentally through an off-resonant single addressing beam inducing Stark-shifts between the on-resonant parallel single qubit rotations executing the *k*-copy twirling. The single addressing beam is detuned from the qubit resonance by -15 MHz, and the induced Stark shift with beam power, γ , is calibrated using an RPE sequence (see section ??), with results shown in figure 1.11 (c) for both the left and right qubit. From the frequency shift versus

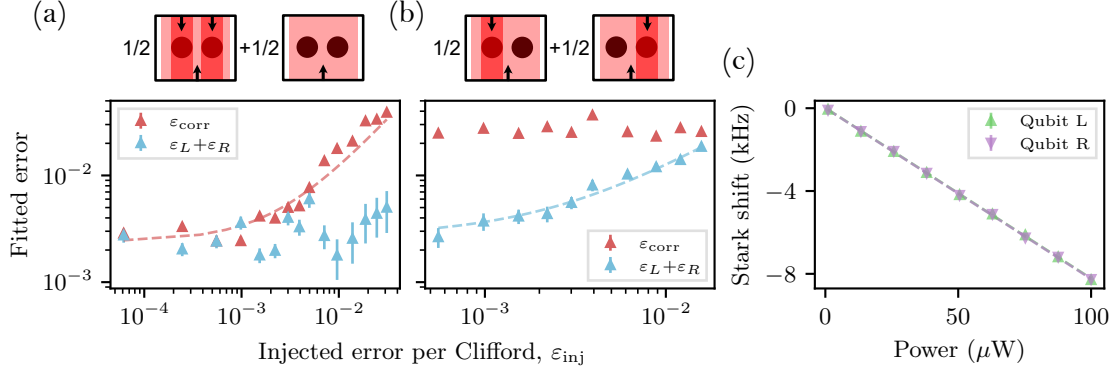


Figure 1.11: Correlated and uncorrelated errors are injected via a single-addressed Stark-shift beam, and quantified through fitting the correlated success populations. The four addressing configurations are depicted visually above the plot. The wide beam signifies the on-resonance global drive which executes the parallel single-qubit rotations, while the narrow beams represent the off-resonant Stark shift beams addressing single-qubits (black circles). In (a) the errors are correlated between qubits, while panel (b) shows the extracted errors after independent errors are injected. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. Panel (c) shows the calibration of Stark shift against total beam power used to infer the injected error.

power, we extract $\gamma_L = 8.24(5) \times 10^7 \text{Hz W}^{-1}$, $\gamma_R = 8.25(4) \times 10^7 \text{Hz W}^{-1}$ for the left and right qubit respectively. The injected error, ε_{inj} , is then inferred by,

$$\varepsilon_{\text{inj}} = \frac{\kappa}{3} \phi^2, \quad (1.77)$$

where the phase error, given in radians is, $\phi = P\tau\gamma$, with P being the power in the single addressing beam, and τ the duration of the Stark-shift pulse. In these experiments, the duration of the Stark-shift pulse is fixed to $\tau = 20 \mu\text{s}$, while the total power of the beam is varied to supply a known error.

Two experimental sequences were run, the first, shown in figure 1.11(a), implements stochastic *common noise* through applying either two spots shifting both ions, or no Stark shift, with a 50 : 50 chance after each Clifford gate. The second experiment, shown in (b), applies local errors through a single spot either addressing the left or right qubit, again with 50 : 50 chance after each Clifford gate. These two arrangements are shown schematically in the upper panels. To ensure correlated errors dominate even with injected local errors, an additional constant correlated error offset of $\varepsilon_{\text{corr},0} \approx 2 \times 10^{-2}$ is applied through random phase fluctuations,

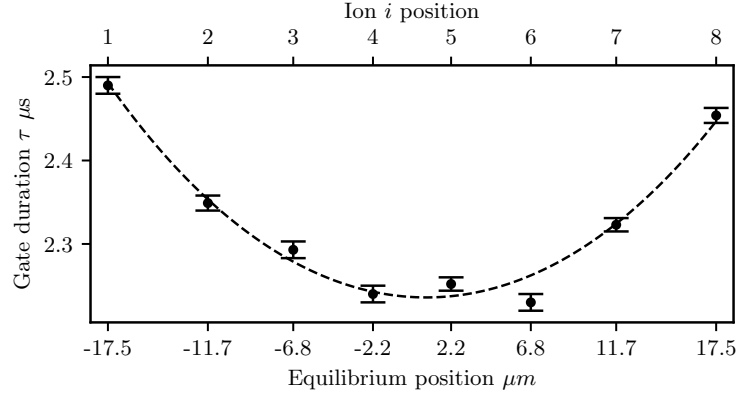


Figure 1.12: Variation of π -pulse gate duration along eight ion chain due to the Gaussian waist of the global addressing beam. Ion labels, i , are given on the upper axis, and equilibrium position given on the lower.

as described in section 1.6.4. This reduces the number of samples required to accurately estimate the correlated and uncorrelated error contributions, which is of practical importance in collecting data in reasonable time (for reference, this data was collected over ~ 3 hours). It can be seen that $\varepsilon_{\text{corr}}$ increases only when injecting *common noise*, and remains constant when local errors are injected. Dashed lines are simple linear fits given by $\varepsilon_{\text{fit}} = \varepsilon_0 + \varepsilon_{\text{inj}}$, where ε_{fit} are the extracted errors, ε_0 is some residual error baseline. These straight line fits show good agreement between measured and injected error when accounting for some fit baseline error rates.

1.6.5 Scaling to a multi-ion register

Pairwise *k*-copy RB extends simply to > 2 qubits: correlated populations are fit to equation 1.36 independently for each qubit pair. For a register of k qubits, all $k(k-1)/2$ pairwise outcomes are extracted from the same shot record — no additional trials on the quantum register are required, although classical post-processing scales with the number of pair combinations. The protocol itself requires only synchronised application of identical single-qubit Clifford sequences and spatially resolved readout.

This scaling is demonstrated on an eight-ion chain, with parallel single-qubit rotations driven by an elliptical 729-nm beam. We expect to observe structured

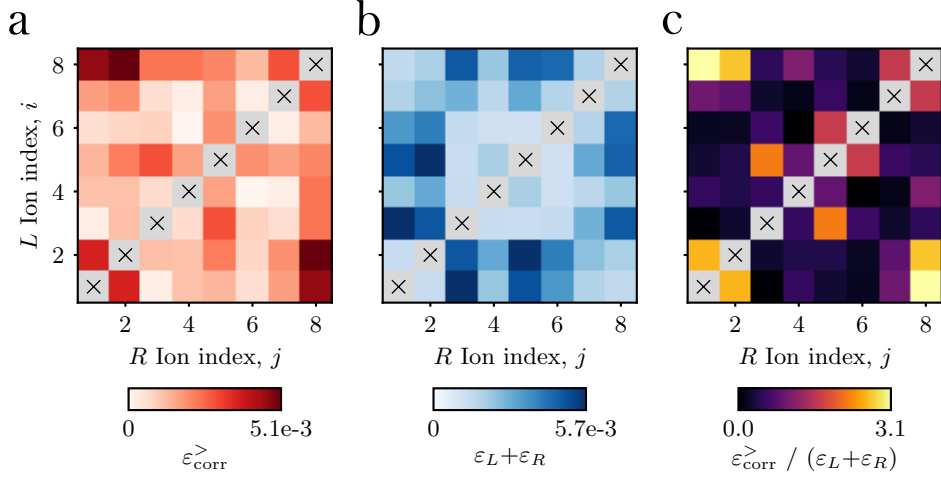


Figure 1.13: Correlated and local error per Cliffords measured for all qubit pairs after applying *k*-copy RB to an 8 ion chain. Noise is dominated by position dependent duration of $R_x(\pi/2)$ gates consistent with inhomogeneous illumination by the global drive. Panel (a) displays the extracted excess correlated errors, $\varepsilon_{\text{corr}}^{(i,j)}$ for each qubit pair, (i, j) . Panel (b) displays the sum of both independent depolarising errors, $(\varepsilon_L + \varepsilon_R)^{(i,j)}$. Panel (c) displays the ratio of excess correlated noise to the sum of local errors given in (b). This plot highlights the outer qubits are dominated by spatially correlated errors.

errors due to the Gaussian intensity profile of the beam. To verify the expected gate duration dependence due to the finite beam, we run parallel gate duration calibration experiments (see section ?? for details). The gate duration, τ , as a function of qubit position in the beam is expected to follow,

$$\tau(r) \propto \frac{1}{\sqrt{I(r)}}, \quad (1.78)$$

where $I(r)$ is a Gaussian distribution, $I = a \exp\left(\frac{-(r-r_0)^2}{2c^2}\right)$. Figure 1.12 shows the measured π -pulse gate durations for each ion, with the fit given by the dashed line. From this, we find an expected major-axis waist radius of 27.7(9) μm aligned to the ion chain. The qubit spacing, on the lower axis of the calibration plot, is inferred through the axial confinement of 0.5 MHz using equilibrium positions of a linear chain given in [47]. At this confinement, the chain extent is 35 μm , giving an expected Rabi-frequency difference of 10.3(6)% between the outer and inner ions. We calibrate global single-qubit gates using the inner ion, $i = 4$, leading to an

expected per Clifford error of (using fidelity definition from appendix A.5),

$$\varepsilon_{c,max} = \frac{2}{3}\kappa \left(\alpha \frac{\pi}{4}\right)^2 = 5.08(2) \times 10^{-3}, \quad (1.79)$$

where, here the Clifford decomposition used gives $\kappa = 1.167$, and the fractional Rabi frequency difference is given by the above calibration curve, $\alpha = 0.103(6)$.

As with the single qubit Stark-shift dataset, we ensure that correlated errors dominate to improve the fitting sensitivity. This is done through a fixed phase noise error corresponding to $\varepsilon_{corr,0} = 1.3 \times 10^{-2}$. In the following data, we subtract this baseline offset to highlight the excess correlated noise due to the beam structure,

$$\varepsilon_{corr}^> = \varepsilon_{corr} - \varepsilon_{corr,0}.$$

Figure 1.13 shows the extracted pairwise correlations. The excess correlated error is shown in (a), with the largest values seen for the outer ion pairs, agreeing with the expected maximum error of $\varepsilon_{corr,max}^> = \varepsilon_{c,max}$. Panel (b) shows the sum of the local depolarisation error per Cliffords on both qubits, with the largest values now corresponding to pairs consisting of one central ion, and one outer ion. We plot the ratio of excess correlated error and depolarisation errors in panel (c) to further highlight the error distribution due to the beam shape.

1.7 Outlook

Distill what learnt. *k*-copy RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. The symmetry enforced by twirling over the diagonal single-qubit Clifford representation $Cl_1^{\otimes k}$ (section 1.3.2), causes structured errors to drive the decay of correlated populations into invariant subspaces. Observing transient populations in these invariant subspaces allows for error characterisation without full process reconstruction. Given a specified error model, here focussing on common unitary errors due to fluctuating global fields, these correlated decays can be used to directly quantify the magnitude of the individual error channels contributing to the total error per Clifford that is typically measured in standard RB.

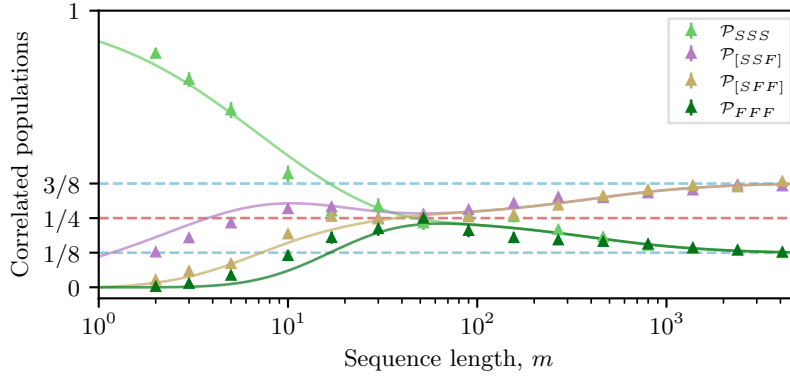


Figure 1.14: Experimental data and analytic fit for 3-qubits subject to correlated random-unitary single-qubit noise injected via random phase jitter ($\sigma_\phi = 0.06$ turns). $\mathcal{P}_{[SSF]}$ and $\mathcal{P}_{[SFF]}$ represent the sum of the permutations of single- and two-qubit failures respectively. Explicitly, $\mathcal{P}_{[SSF]} = \mathcal{P}_{SSS} + \mathcal{P}_{SFS} + \mathcal{P}_{FSS}$.

In this chapter we proposed and experimentally demonstrated this method applied on up to an eight-qubit register through pairwise comparisons (section 1.6.5). This approach permits highly efficient parallel data collection: the time required on the device is independent of qubit number, and the protocol requires exclusively local operations followed by classical post-processing of correlated survival rates.

reiterate relevance.

1.7.1 Limitations

add parameter regime that is hard to fit. add the need for explicit channels? This chapter has analysed structured errors on two qubits, extending to more qubits only through pairwise comparisons. The current pairwise protocol described here does not determine correlated errors between > 2 -qubits. These higher weight errors may be particularly detrimental to error correction circuits where the distance of the code, d , is less than the correlated error channel weight [**<empty citation>**]**citation needed**, therefore these errors would be advantageous to characterise. We expect the symmetry resolved dynamics revealed by pairwise *k*-copy RB to generalise to higher weight errors, however, without prior work on the sub-representation structure of the single-qubit Clifford group applied

diagonally to > 2 qubits, extending the analysis for *k-copy* RB is challenging. The number of relevant irreps does not increase, however, the multiplicity does increase dramatically. Some headway can be made if we consider the *common noise* error mode discussed throughout this chapter. A simple relation can be found for the asymptotic states, namely, under *common noise* the state is driven to the maximally mixed state on the symmetric subspace, as this is the $SU(2)$ -invariant state on the symmetric irrep. This state is given by,

$$\rho_{\text{sym}} = \frac{1}{k+1} \sum_{l=0}^k |D_l^k\rangle \langle D_l^k|, \quad (1.80)$$

where $|D_l^k\rangle$ is the Dicke state for l excitations (here successful recoveries of target state), out of k -qubits. From this expression we see that an asymptote of $\mathcal{P}_{SS\dots S}|_{m \rightarrow \infty} = 1/(k+1)$ is expected under *common noise*, while the maximally mixed state under local depolarisation leads to the familiar asymptote of $\mathcal{P}_{SS\dots S}|_{m \rightarrow \infty} = 1/(2^k)$. This result suggests that the presence of *common noise* affecting many qubits may be simply observed, however, without the explicit representation structure, it cannot be easily quantified.

We confirm this asymptote experimentally with data shown in figure 1.14. For three-qubits, we leverage symbolic computation to diagonalise the resulting PTM representation of *common noise* and recover the analytic decay used in this fit. This direct diagonalisation approach is possible for this small system, and with the highly structured *common noise* error channel, however, for a full treatment of k -body error under $Cl_1^{\otimes k}$ twirling, we suspect the sub-representation structure will need to be found.

1.7.2 Future work

One avenue for future work is captured by the above limitations section: scaling the analysis to multi-body errors through studying the sub-representation structure of the single qubit Clifford group applied diagonally to k -qubits. We propose two further dimensions in which to extend the *k-copy* analysis:

First, the subsystem on which the local twirling is applied could be extended to

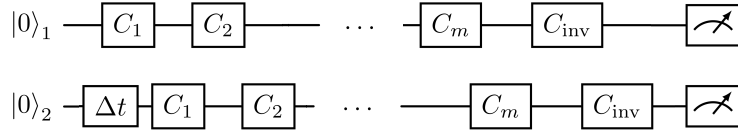


Figure 1.15: Circuit for probing time-dependent environmental errors by introducing a delay between random sequence execution on parallel copies in *k*-copy RB.

> 1 -qubit. As an example, the two-qubit Clifford group [48] or the SLeRB [35] protocol could be applied in parallel to k -copies of 2-qubit subsystems. This would reveal correlated errors and leakage that reduce the fidelity of two-qubit entangling gates.

Second, we may utilise this method to probe correlations in time. As the protocol uses exclusively single-qubit rotations, with no explicit coupling made between qubits by the ideal circuit, we could apply some time delay, Δt , between the parallel gate sequences. A proposed circuit is shown in figure 1.15 for the two-copy case. In the case that Δt is much longer than the random sequence length, we are effectively probing the stability of the circuit over time (albeit on parallel subsystems) — specifically we are looking for deviations in circuit fidelity due to the external context in which the circuit is executed. This method is similar to the context dependent tests studied in reference [31]. However, as we execute parallel sequences on two copies, we can also set Δt to be significantly shorter than the random sequence length. In this regime, we could, for example, probe the frequency and coherence of noise on the common drive field used to execute the parallel gates. This would provide evidence and quantify violations of the Markovian QIP model, without full process tomography, as discussed in section 1.1.

2

Towards Hybrid System Control and Quantum Error Mitigation

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Having established and characterised the experimental apparatus and demonstrated its application to *k-copy* randomised benchmarking, this chapter considers future research goals utilising the multi-qubit arrays enabled through the *NPL* trap (§ ??), the single addressing system (§ ??), and the developed control over both the discrete qubits, as well as the continuous motional modes.

In § 2.1 we establish a set of unitary operations required for universal continuous-variable (CV) control of the ion motion and discuss the extensions to hybrid spin-oscillator systems. Following on from recent progress in spin-conditioned

bosonic operations demonstrated within our group [49–53], we are interested in developing the required addressing capabilities to scale to multiple qubits and motional modes. We outline the method we use to generate such spin-oscillator interactions experimentally, and demonstrate the generation of a squeezed state on the new apparatus. We then outline the next steps required for demonstrating the remaining single- and two-mode interactions, and the possible experiments this will enable.

The second half of this chapter considers an error mitigation technique applicable to both discrete and continuous systems. Developing this, and associated methods on our trapped ion platform will allow substantial reductions of certain errors, with significantly less resources required than full error correction. We give a short introduction to the purpose of quantum error mitigation, followed by an outline of virtual distillation [54–56] — the specific method considered here. We provide an initial demonstrations of this protocol applied to a discrete-variable only system, and explore the performance under both local depolarising and correlated noise due to a common fluctuating control field (see § 1.3.3 for details).

We conclude this chapter with a discussion on the application of virtual distillation to hybrid systems, presenting a scalable method within reach of our apparatus.

2.1 Hybrid Spin-Oscillator Control in Trapped Ions

CV quantum computation [57] offers an alternative strategy for encoding quantum information, utilising quantum harmonic oscillators, termed *qumodes*, as the fundamental quantum object, rather than discrete variable (DV) systems, e.g. the two-level qubit. A single ideal qumode possesses an infinite dimensional Hilbert space, allowing for high (in this case arbitrarily high) density encoding of information in physical information carriers. This opens an alternative avenue for scaling QIPs, requiring dramatically less physical carriers compared to qubit-only systems. The inclusion of both CV and DV subsystems, termed the hybrid model [58], retains the convenient control and readout of discrete qubits while providing the larger state

space and native interactions available to qumodes. For example, reference [59] considers a factoring algorithm with the required physical resources independent of the number to be factored. Remarkably, this scheme requires only a single qubit and three oscillators, however requires prohibitively large energy states of the oscillator. Experimentally, we will not be able to realise arbitrarily high energy oscillator states, but it is still an open question of what information density is possible in accessible engineered quantum oscillator.

Hybrid spin-oscillator computation has been studied in the context of superconducting cavity-QED [60–64], neutral atom in optical tweezers [65–67], and in the trapped-ion platform [68–70]. We shall focus on the trapped-ion implementation, where the DV system is the collection of qubits embedded in the electron energy levels, and the qumodes are the shared secular motional modes of the ion crystal within the trapping pseudopotential.

Trapped ions benefit from multiple native methods to drive, and couple qubit and qumode: The qumodes may be directly driven through parametric trap potential modulation [71–73], or through near mode-resonance RF fields; the qubit may be driven either by optical fields (in the case of optical qubit encoding), or through microwaves (in the case of ground state encoding) resonant with the energy-state transition; the qubit and qumode may be coupled through qubit-state dependent potential shifts [74, 75], or through optical fields tuned near to motional-mode frequency sidebands of the qubit-carrier transition, for example through the Jaynes-Cummings interaction (§ ??), or the Mølmer-Sørensen (MS) SDF [76]. Taken together, these native control and coupling mechanisms, combined with the long coherence times of both the qubit and qumode, make the trapped-ion platform well-suited for exploring hybrid CV–DV quantum computation.

The apparatus developed in this thesis has demonstrated good qubit and radial motional coherence times, provides access to multiple qubits and qumodes, and includes a single-addressing system for selectively driving their coupling. These capabilities make it well-placed for studying methods for scaling hybrid-system control.

Interaction name	Oscillator H	Gauss.	N_{osc}	Two-SDF method		
				m	$\hat{\sigma}$	Ω
Displacement	$ae^{i\phi} + a^\dagger e^{-i\phi}$	✓	1	-	$\hat{\sigma}_x$	Ω_α
Parity	$aa^\dagger e^{i\phi} + a^\dagger a e^{-i\phi}$	✓	1	+1	$\hat{\sigma}_z$	Ω_2
Beam splitter	$ab^\dagger e^{i\phi} + a^\dagger b e^{-i\phi}$	✓	2	+1	$\hat{\sigma}_z$	Ω_2
Squeezing	$a^2 e^{i\phi} + a^{\dagger 2} e^{-i\phi}$	✓	1	-1	$\hat{\sigma}_z$	Ω_2
Trisqueezing	$a^3 e^{i\phi} + a^{\dagger 3} e^{-i\phi}$	✗	1	-2	$\hat{\sigma}_x$	Ω_3

Table 2.1: A suggested set of unitary interactions to compose a universal gate set for continuous variable computation. Using the protocol presented in reference [68], here termed the “Two-SDF” method, we show the required relative detuning setting, m , and the resulting spin conditioning, $\hat{\sigma}$, and coupling strength Ω of the resulting interaction. Ω_α is the typical SDF strength given by equation ???. The other coupling strengths are found via the 2nd and 3rd terms of the Magnus expansion: $\Omega_2 = 2\Omega_\alpha\Omega_{\alpha'}/\Delta$, $\Omega_3 = 2\Omega_\alpha^2\Omega_{\alpha'}/\Delta^2$.

2.1.1 Theory

Gateset for a Hybrid System

For any computing paradigm, we must establish a set of interactions required for universality. In DV-only systems, quantum universality may be realised by a finite set of Clifford operations plus one non-Clifford gate [77, 78]. CV-only universality can analogously be realised by a finite set of Gaussian operations plus one non-Gaussian gate [57, 79]. The term Gaussian is understood in the position-momentum representation of the qumode, where the bare oscillator Hamiltonian is given by,

$$H_{\text{osc},0} = \hbar\omega_{\text{osc}}(\hat{x}^2 + \hat{p}^2), \quad (2.1)$$

with quadratures,

$$\hat{x} = \frac{1}{2}(a + a^\dagger), \quad (2.2)$$

$$\hat{p} = \frac{-i}{2}(a - a^\dagger), \quad (2.3)$$

where $a^\dagger$, a are the creation and annihilation ladder operators of the mode with frequency, ω_{osc} . Gaussian operators are then the set of interactions containing at most quadratic terms in $\hat{x}$ and $\hat{p}$, therefore mapping Gaussian input states to Gaussian output states.

The condition for universality of a hybrid CV-DV is somewhat ambiguous as both

the CV and DV subsystems are themselves sufficient for universality, however this presents an opportunity: the freedom in selecting the chosen gate-set for universality, combining both Clifford, non-Clifford, Gaussian, non-Gaussian, and spin-oscillator interactions together, allows for selecting efficient, native, and simply accessed interactions of the physical platform. We consider the over-complete instruction set presented in Table 2.1, consisting of spin-conditioned Gaussian and non-Gaussian interactions. The interactions on the bosonic subsystems are themselves universal, but the spin-conditioning provides a bridge between CV and DV subsystems, allowing for readout and control to be implemented on ancilla qubits. Specifically, we consider a cubic CV instruction set [57, 60, 80], taking the Gaussian operations and supplementing with a non-Gaussian “tri-squeezing” gate, with Hamiltonian,

$$H_T \propto a^3 + a^{\dagger 3}. \quad (2.4)$$

Spin-conditioned tri-squeezing has been shown previously in our group [49], utilising the protocol given in reference [68], here termed the “two-SDF” method.

The remaining non-linear Gaussian operations; squeezing, parity, and beam-splitter, may be generated through the same “two-SDF” mechanism, which we shall detail in the following section, and have also been previously demonstrated [49, 51, 52, 81]. The displacement operation can be realised either spin-independently by RF fields near resonance to the motional-mode frequencies (§ ??), or spin-conditioned using various SDF implementations, such as by the MS method using bichromatic fields (§ ??), or by static Stark-shift gradients due to an optical tweezer [75].

As highlighted in this section, the required unitaries for universal hybrid CV-DV control may all be accessed by a common protocol: The “two-SDF” method. This greatly simplifies experimental hardware and control complexity. We shall give an overview of this method in the following section, and provide a practical route and demonstration utilising four optical frequencies with fixed relative detunings and phases.

- Do we need to define each interaction, sqz, parity, bs?
- Could give flow fields for single-mode operators (could be a nice figure, and I like that representation).

Gate Implementation

To implement the unitaries given in table 2.1, and to retain spin-dependent control of these interactions to fully utilising the large hybrid state space, we use the “two-SDF” method proposed in reference [68]. The non-linear bosonic interactions are effectively driven through applying two linear spin-dependent forces with non-commuting spin conditioning, simultaneously on a single qubit. The two simultaneous SDFs, with Hamiltonians and parameters labelled by subscripts α , and β , are applied off-resonantly relative to motional modes with creation operators $\hat{a}_\alpha^\dagger$ and $\hat{a}_\beta^\dagger$, respectively. The resulting Hamiltonian is given generally by,

$$\hat{H}_{\text{2SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (2.5)$$

$$= \frac{\hbar\Omega_\alpha}{2}\hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (2.6)$$

$$+ \frac{\hbar\Omega_\beta}{2}\hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right), \quad (2.7)$$

where the individual SDF strengths are Ω_i , the detuning from each mode is Δ_i , the motional phase is ϕ_i , and we ensure the spin-conditioning anticommute $\{\hat{\sigma}_\alpha, \hat{\sigma}_\beta\} = 0$. The unitary dynamics of $\hat{H}_{\text{2SDF}}$ may be evaluated by the Magnus expansion [82], with details given in references [51, 52, 68], and in appendix A.8. To lend some intuition on the resulting physics, the Magnus expansion expresses the generator of the unitary evolution as an infinite series of terms, with successive terms proportional to increasingly higher-order nested commutators of the Hamiltonian evaluated at different times. Due to the non-commuting spin conditioning, this series does not truncate, leading to arbitrarily high non-linear bosonic interactions effectively driven via the application of only two linear SDFs.

To utilise specific non-linear terms within this infinite series, matching the desired interactions given in table 2.1, we constrain the relative detunings of the two SDFs

to $\Delta_\beta = m\Delta_\alpha$, where m is some integer given in the table. This immediately brings components of the Magnus terms matching the desired non-linear $\hat{H}_{\text{osc}}$ into resonance. Undesired higher-order terms also brought into resonance may be arbitrarily suppressed as the effective coupling of the n th Magnus term is $\Omega_{\text{eff}} \propto (\Omega_\alpha/\Delta)^{n+1}$ (assuming $\Omega_\alpha = \Omega_\beta$). The remaining error terms are far off-resonant, and may also be further suppressed by ensuring $\Delta \gg \Omega_\alpha\Omega_\beta$, and utilising appropriate smooth amplitude and phase envelopes for the pulses.

In this regime, the unitary dynamics of $\hat{H}_{\text{2SDF}}$ are well approximated by the evolution of time-independent effective Hamiltonians, which for Gaussian operations ($|m| = 1$) are of the form,

$$\hat{H}_{\text{eff,G}} = \frac{\hbar\Omega_\alpha\Omega_\beta}{2\Delta}\epsilon_{\alpha\beta\gamma}\hat{\sigma}_\gamma(ab^{(\dagger)}ie^{i\phi} + h.c.), \quad (2.8)$$

where $\epsilon_{\alpha\beta\gamma}$ is the Levi-Civita symbols. If $a_\alpha^\dagger = a_\beta^\dagger$, the SDFs are detuned from the same motional mode, leading to squeezing, or parity effective interactions, while acting on two-modes leads to the beam splitter interaction. For the tri-squeezing interaction ($m = -2$) the effective Hamiltonian is,

$$\hat{H}_{\text{eff,T}} = \frac{\hbar\Omega_\alpha^2\Omega_\beta}{4\Delta^2}\hat{\sigma}_\beta(a^3e^{i\phi} + a^{\dagger 3}e^{-i\phi}), \quad (2.9)$$

where we have set $a_\alpha^\dagger = a_\beta^\dagger$, as we act on a single mode.

Frequency and spin-conditioning control of two SDFs therefore allow for the generation of the entire proposed hybrid CV-DV gate-set. After a discussion on how we can characterise the oscillator state, we shall present the experimental scheme for generating these interactions, and demonstrate the creation of a squeezed state by this method.

Readout

With the above methods we may generate interesting non-classical bosonic states, now we provide a method for characterising them.

In contrast to a two-level system, the state of an oscillator may not be directly readout through state dependent fluorescence. This is due to the much larger

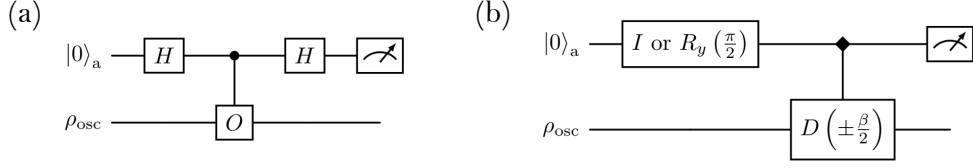


Figure 2.1: State tomography for a single mode through measurement of an ancilla qubit. (a) The Hadamard test method for measuring the expectation value of an arbitrary operator, $\hat{O}$, applied to the oscillator state, utilising an ancilliary qubit. (b) A Hadamard test circuit for measuring the characteristic function, $\chi(\beta)$, of the oscillator using native single qubit rotations and σ_x -conditioned spin-dependent forces (symmetric control represented by the diamond symbol) [83].

state-space of the qumode, and the technical difficulty in direct motional readout on the ion-trap platform. To solve these issues, a modular readout scheme is used, involving indirect readout of the oscillator state via a discrete collection of controlled unitaries subject to an ancillary qubit, and computational basis measurements of that ancilla.

The information is mapped to the ancilla by a Hadamard test [84], shown in figure 2.1 (a). Here the expectation of value due to the operator $\hat{O}$ on the oscillator, $\langle \hat{O} \rangle = \text{Tr}(\hat{O}\rho_{\text{osc}})$, is recovered through exclusive measurements of the ancilla qubit after qumode-dependent phase “kick-backs” are applied. The Hadamard test has the convenient relation between computational basis measurements of the ancilla, and the applied operator, $\langle \hat{\sigma}_z \rangle = \text{Re}[\text{Tr}(\hat{O}\rho_{\text{osc}})]$.

Full state tomography may now be formulated by repeat measurements of the conditional displacement operation, $\hat{O} = e^{-i\theta}\hat{D}(\beta)$, to recover the characteristic function, defined as $\chi = \text{Tr}(D(\beta)\rho_{\text{osc}})$, β is a complex parameter with displacement magnitude, $|\beta|$, and displacement axis, $\arg(\beta)$. Final computational state measurements of the Hadamard test circuit then yield,

$$\langle \sigma_z \rangle = \text{Re}[e^{-i\theta}\text{Tr}(D(\beta)\rho_{\text{osc}})], \quad (2.10)$$

$$= \cos(\theta)\text{Re}[\chi] + \sin(\theta)\text{Im}[\chi], \quad (2.11)$$

where θ may be set to 0 or $\pi/2$ to recover the real or imaginary part of the characteristic function respectively. Each measurement provides only 1-bit of

information on the (continuous) oscillator state, thus to approximate a useful reconstruction, many samples with a range of β values must be collected.

Once the characteristic function has been measure, it can be converted to other familiar representations of the oscillator, such as in the Fock-state basis, or Wigner representation [85–87]. In this work, we shall keep to the conveniently measured characteristic function.

To give two concrete example, the vacuum state characteristic function is the circularly symmetric Gaussian state,

$$\chi_0 = \exp\left(\frac{-|\beta|^2}{2}\right), \quad (2.12)$$

while the squeezed state, with squeezing parameter $\xi = re^{i\phi}$, breaks this symmetry due to a contraction on one axis, and a corresponding expansion on the other [88],

$$\chi_s = \exp\left(\frac{-|\beta|^2}{2} \left(e^{2r} \cos^2(\phi') + e^{-2r} \sin^2(\phi')\right)\right), \quad (2.13)$$

where ϕ' is the angle between squeezing axis and probe axis, $\phi' = \arg(\beta) - \frac{\phi}{2}$.

Experimentally the $\hat{\sigma}_x$ conditioned spin-dependent force are simply accessed by the bichromatic field method discussed in § ?? . The Hadamard test circuit is decomposed to native gates, shown in figure 2.1 (b), first described in reference [83]. The proceeding I or $R_y(\pi/2)$ set the quadrature to measure; $\text{Re}(\chi)$ or $\text{Im}(\chi)$ respectively. This circuit provides a practical method for characterising a single qumode, which we shall use in the following experimental demonstration. We note, similar to the constraints of state characterisation discussed in § 1.1, full tomographic reconstruction is intractable for multiple-qumodes due to the large state-space, even with reasonably truncated oscillator representations (e.g. imposing a Fock-state cut-off). In the case of characterising systems consisting of multiple qubits and qumodes, either reduced tomography [89], or simplified metrics of performance [90] must be used.

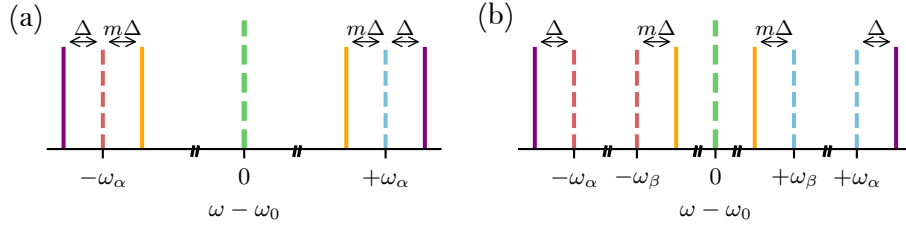


Figure 2.2: The quadchromatic field required for generating non-linear motional interactions via the two-SDF method. (a) For single-mode interactions, both SDF fields are detuned around the sidebands of the same motional mode, ω_α . Integer m is varied to select the desired interaction. (b) For two-mode interactions, each SDF is detuned from a different motional mode.

2.1.2 Experimental Demonstration

In our group, we have previously demonstrated higher order squeezing [49], superpositions of non-linear bosonic states [50], beam-splitter [81], and two-mode squeezing interactions. However, due to the constraints of the previous experimental apparatus we were limited to global addressing fields, preventing the application of these hybrid interactions to more than two qubits. Here we demonstrate the creation and characterisation of a squeezed state on the newly constructed apparatus, described in chapter ??, utilising the single-qubit addressing system. Single addressing both increases the power efficiency of the interactions, and allows the control to select which qubit is used to couple to the motion — this provides a scalable route to generating hybrid interactions on larger qubit, and qumode registers.

Two Spin Dependent Forces

The individual SDFs are generated by the same bichromatic field method described in § ?? — i.e. simultaneous off-resonant blue- (BSB) and red-sidebands (RSB) of motional mode ω_α are applied. As such, we require four optical fields with, frequency, phase, and amplitude control. The four required frequencies of the “quadchromatic” field are shown schematically in figure 2.2 with respect to the carrier frequency, ω_0 . Here, purple lines correspond to SDF_α , and orange lines to SDF_β . In this example we show the SDFs acting on a shared motional mode,

but in general they may act on different modes. Similar to the generation of the bichromatic field, we may produce this quadchromatic field through the application of four RF tones onto a single-pass (SP) AOM. These tones are generated using an arbitrary waveform generator (AWG)¹. We cannot control the absolute optical phase at the ion, however we can set relative phases between the four fields by applying phase offsets between the RF tones with respect to some fixed time point. As discussed in § ??, the spin phase of the SDFs is set by the absolute phase of the RSB and BSB, $\hat{\sigma}_\phi = \cos(\phi_s)\hat{\sigma}_x + \sin(\phi_s)\hat{\sigma}_y$. To ensure we saturate the non-commutativity of the spin conditioning of the two SDFs, we require $\pi/2$ phase difference between the two spin phases,

$$\phi_{\alpha,s} - \phi_{\beta,s} = \frac{\phi_{\alpha,\text{RSB}} + \phi_{\alpha,\text{BSB}}}{2} - \frac{\phi_{\beta,\text{RSB}} + \phi_{\beta,\text{BSB}}}{2} = \frac{\pi}{2}. \quad (2.14)$$

In the following demonstration we set $\phi_{\alpha,\text{RSB}} = \phi_{\alpha,\text{BSB}} = \pi/2$, $\phi_{\beta,\text{RSB}} = \phi_{\beta,\text{BSB}} = 0$, however we note that for composing multiple hybrid operations into a circuit, the resulting motional phase of the SDFs can lead to further constraints on the chosen RSB and BSB phases.

The intensity of the quadchromatic field is ramped on and off with a rise envelope of $\sin^2(\pi t/(2t_{\text{ramp}}))$ applied via the AWG, where t_{ramp} is the ramp duration. This ramp suppresses excess off-resonant excitations of the A_1 phase loop term given in the previous section, and so we must select its duration based on the characteristic scale, $\frac{1}{\Delta}$.

We aim to balance the power in each of the 4 tones. This may be calibrated using an optical pick-off before addressing the ion, directed onto a fast photodiode, and measuring the beatnote contrast when applying any pair of tones. In the following squeezing demonstration, we balance the tones comprising the individual SDFs, but do not explicitly balance the power between SDFs, as the desired interaction scales as $\Omega_\alpha\Omega_\beta$, any imbalance leads to non-optimal interaction efficiency, but unlike imbalance of the RSB and BSB, does not increase coherent error terms. We observe that the duty cycle of the SP AOM can dramatically shift the power imbalance

¹*Sinara Phaser* board

between all four tones, and so we take care ensuring that the total RMS power to drive one-, two-, and four-tones is the same as for the idle (no optical pulse) setting of the SP AOM, thus the only transient power drop is that of the ramp sequence.

Squeezing

- **potential plot: check ramp is sufficient by looking at σ_z dependence. If want ramp have rid 120238, 119279**

We demonstrate the effective squeezing interaction on a single ion and radial motional mode. We are limited to using the radial modes due to the geometry single addressing system with respect to the trap. We use the upper radial mode with frequency, $\omega_{upper} = 2.87$ MHz, and motional coherence $\tau_{upper} = 19(1)$ **rid 135596** given in § ???. The initial near-vacuum motional state is prepared by Doppler and sideband cooling, with a typical thermal fock state distribution with mean $\bar{n} = 0.03(1)$, as described in § ??. The spin state is initialised into the ground state by optical pumping, described in § ??. By setting the relative detuning between the SDFs to $m = -1$, and ensuring anti-commuting spin conditioning by setting the relative phases of the quadchromatic field according to equation 2.14, we realise a $\hat{\sigma}_z$ -conditioned squeezing interaction. As the spin-state is prepared in an eigenstate of the spin-conditioned squeezing, we expect squeezing on the qumode, without any change to the qubit state. We use a total quadchromatic beam power of 53 μ W, and set the base detuning, $\Delta = 2\pi$ 50 kHz. We suppress excess excitation of the off-resonant error terms by using a ramp of $t_{ramp} = 40$ μ s.

The state is characterised by the tomography method given in § 2.1.1, with a discretization set to collect a 35×35 grid of displacement values, β , and $N = 300$ repeat samples to reduce shot noise. The full dataset for one of the tomographic scans takes around 1 hour to collect, and so we interleave mode calibration experiments (§ ??) to run every 5 minutes.

The on-resonance splitting SDF used for tomography must have its strength, Ω_{SDF} ,

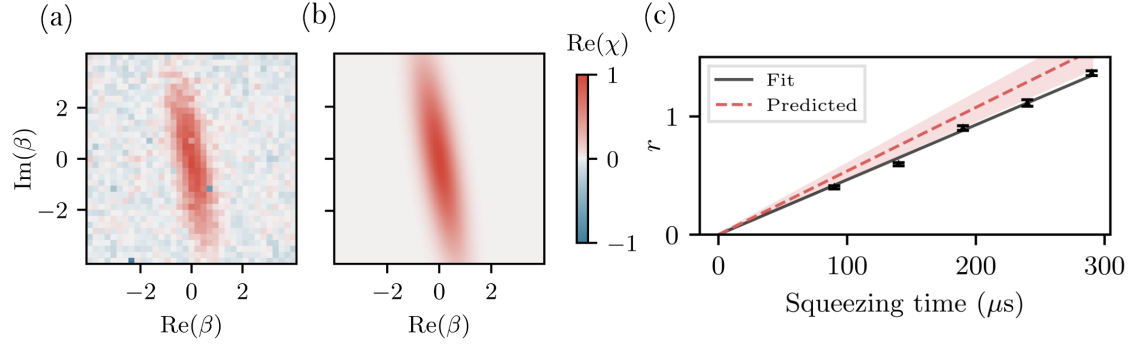


Figure 2.3: Demonstration of the squeezing interaction through the two-SDF method. (a) The characteristic function, $\text{Re}(\chi(\beta))$, of the squeezed state measured using the state tomography method given by figure 2.1. (b) The simulated state taking fit parameters inferred by equation 2.13. (c) The fitted squeezing parameter, r , as a function of two-SDF pulse duration.

calibrated to infer the magnitude of β applied. We use a total bichromatic beam power of 40 μW , and through fitting the $\hat{\sigma}_z$ expectation for the time-dependent splitting of a vacuum state,

$$\langle \hat{\sigma}_z \rangle = e^{-2|\beta|^2}, \quad (2.15)$$

with

$$|\beta| = \frac{\Omega_{\text{split}} t}{2}. \quad (2.16)$$

We find $\Omega_{\text{split}} = 2\pi \cdot 7.2(4)$ kHz.

The pulse sequence for squeeze state generation and characterisation is given in figure ??, and the resulting real part of the characteristic function is shown in figure 2.3 (a). Fitting the data to equation 2.13 we may extract both the squeezing axis, ϕ , and the strength, r , where r is given by,

$$r = |\Omega_{\text{eff},S}|t = \frac{\Omega_\alpha \Omega_\beta}{2} t. \quad (2.17)$$

An example fit is shown in panel (b), with extracted $r = 0.90(2)$, for a quadchromatic FWHM pulse duration of $t_{\text{sqz}} = 190 \mu\text{s}$. Panel (c) displays the squeezing strength as a function of t_{sqz} , with fit values given by the points, and the black line is a linear fit. Assuming that the two SDF strengths are well balanced, $\Omega_\alpha = \Omega_\beta$, and taking the 53 μW used for the quadchromatic field compared to the SDF strength calibration

at 40 μW , we find an expected squeezing strength of $\Omega_{\text{eff},S} = 2\pi \cdot 0.86(9)$ kHz. This is shown as the red dashed line and shaded region. The fitted strength of $\Omega_{\text{eff},S} = 2\pi \cdot 0.74(1)$ kHz is lower than the expected, suggesting that either we are not sufficiently balancing the strengths of the two simultaneous SDFs, or that the power stabilisation routine for 4-tones is inaccurate. Further exploration is needed, including careful quantification of the individual SDF strengths constituting the quadchromatic field. However, this preliminary result confirms that the motional coherence, and stability of the single-addressing system is sufficient to create, and characterise non-classical motional states using the apparatus developed for this thesis.

Comment on heating rate

2.1.3 Outlook

Hybrid systems offer access to large Hilbert spaces, even with modest ion numbers. In this section we outlined a convenient set of native interactions for universal hybrid qubit-qumode gates. We experimentally demonstrate the squeezing interaction generated via the “two-SDF” method of reference [68]. This demonstration replicated previous work using the “two-SDF” method [49], however now shown on the newly built apparatus utilising a single-ion addressing system, and applied to the radial motional modes.

Enabled by the single-addressing, and improved motional coherence times of the new system, we are particularly interested in exploring short hybrid circuits composed of multiple interactions acting on an array of qubits and qumodes. Specifically, we are interested in utilising the hybrid system with increased robustness against errors through the use of quantum error mitigation methods applicable to both spin and oscillator subsystems. We outline one such method in the next section, with a preliminary demonstration on the spin subsystem.

Limitations

Using a single multi-ion chain to access multiple qubits and qumodes for scaling hybrid QC on the trapped-ion platform will be ultimately limited by spectral crowding of the motional modes, i.e. the average frequency spacing between neighbouring modes decreases as the number of accessible modes increases.

The “two-SDF” method requires frequency selectivity to drive bosonic interactions exclusively on the chosen mode. To selectively drive a single mode, the magnitude of the detuning Δ must follow $\Delta \ll \delta\omega$, where $\delta\omega$ is the frequency difference to the nearest neighbouring mode. As discussed in the “two-SDF” method section, the strength of each SDF must be less than the base detuning, $\Omega_\alpha \ll \Delta$, to well approximate the effective non-linear unitary dynamics. As a result of these two requirements, the mode-frequency splitting directly limits the attainable strength of the non-linear interaction in the adiabatic regime, resulting in increasing gate duration with increasing register size.

A possible mitigation would be to rather scale the hybrid system using a collection of fixed-length chains with adequate mode-frequency splitting, and to transport quantum information between these chains either through ion transport whilst maintaining mode and spin coherence [91], generating remote entanglement through photonic interconnects [92], or through inducing collective motional interactions between separate ion-chains [93].

Another mitigation strategy could come from the pulse-shaping approach demonstrated for use in fast two-qubit entangling gates in the non-adiabatic regime [94], i.e. when $\Delta > \delta\omega$. Introducing extra degrees of freedom in the pulse amplitude envelope could allow for driving the desired interaction on the selected modes, while returning all spectator modes to their initial state by the end of the pulse. However, we note that the accumulation of non-global geometric phases, and possible non-commutativity of interactions on shared qubit-oscillator states could prohibit the use of such a pulse-shaping scheme. Ultimately, the desired interaction, mode frequency spectra, and target gate duration will determine the dominant coherent

errors. We leave such an analysis for future work.

2.2 Virtual Distillation

With access to noisy, intermediate-scale quantum information processors, the natural question is whether useful computations may be executed, without requiring full fault-tolerant error correction codes. In this section, we explore a resource-efficient error mitigation technique, known as virtual distillation, that can dramatically suppress the effect of errors on the final measured outcome for expectation value problems.

This section starts with outlining the goals of quantum error mitigation (QEM), with the following section giving details on the virtual distillation method. We then present an experimental demonstration suppressing errors a single qubit system. We finish this section with a discussion on future plans, and the extension of virtual distillation to hybrid CV-DV systems.

2.2.1 Error Mitigation

Noise coupled to a quantum system results in unintended coherent, and incoherent evolution. Deviations of the realised state from the target state are known as errors, and, if left unchecked, lead to the failure of quantum algorithms. This failure can either be due to a reduction of signal to noise ratio, or a bias in the measured expectation value, resulting in misleading conclusions. Ultimately, the goal of quantum error correction (QEC) is to detect and correct these errors through embedding information in larger Hilbert spaces [qec_review]. Full QEC will allow for the execution of arbitrarily long algorithms ensuring errors (of up to a certain distance) do not accumulate over time. However, current QEC methods require large overheads in both qubit number and circuit depth, and are therefore not feasible on current quantum hardware [] [is this fair to say?].

Furthermore, QEC is formulated for digital “gate-based” quantum computers, which although promise to be universal, pay for that privilege in often inefficient decompositions of target unitaries into a discrete set of gates. This poses an opportunity for near term devices to show quantum advantage using either short depth, uncorrected circuits, or analog quantum simulators consisting of tailored, often continuous, unitary evolution [nisq_review]. Particularly, phrasing the problem in terms of expectation values of an observable on a physical final state, rather than a sequence of measurements on a set of logical states, allows for a more efficient use of resources. As a tactile example, consider the problem of finding the ground state energy of a Hamiltonian. This can be done by preparing a trial state, and directly measuring the expectation value of the Hamiltonian on that state, building up statistics over many samples [variational_qc_review].

However, the impact of errors on uncorrected computation must be carefully managed. Uncertainty of the approximated expectation value due to averaging over a finite number of discrete samples, known as shot-noise, may be arbitrarily reduced by increasing the samples, N . However, due to imperfect execution of a given circuit, and coupling to the environment, there will be some systematic shift of the estimated average value ideal value in the noise-less limit. This shift, or bias, of the approximated observable may be reduced by the application of quantum error mitigation (QEM) techniques.

Reference [3] defines QEM as “... algorithmic schemes that reduce the noise-induced bias in the expectation value by postprocessing outputs from an ensemble of circuit runs, using circuits at the same noise level as the original unmitigated circuit or above.” This highlights the main distinction to QEC protocols: the physical noisy state is never corrected (often it is in fact degraded through the use of QEM) — rather the output from the noisy circuits is manipulated and postprocessed in such a way that we *effectively* recover an expectation value with suppressed deviation from the ideal value.

rethink narrative here

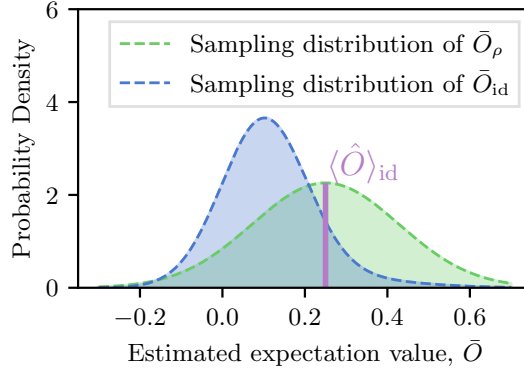


Figure 2.4: **Integrate figure into main text.** $\rho = 0.1\rho_{\text{id}} + 0.9\rho_{\text{err}}$. Blue curve is PDF for N shots, green is PDF for ideal shots, with $N_{\text{id}} = 0.1N$.

A schematic explanation of the goal of quantum error mitigation. Deviations of the measured expectation value of an operator $\hat{O}$ on a noisy state, ρ , and on the ideal state ρ_{id} may be separated into components originating from shot noise, and from an underlying bias. Shot noise sets the variance of the distribution of estimated expectation values, while the underlying bias shifts the center of the distribution. Error mitigation reduces bias at the cost of increasing variance. Here we demonstrate this by the ideal QEM method introduced in § 2.2.1. The underlying noisy state is given by $\rho = (1 - p)\rho_{\text{id}} + p\rho_{\text{err}}$. N shots are taken measuring $\text{Tr}(\hat{O}\rho)$. The ideal QEM flags and discards shots originating from ρ_{err} , leaving $(1 - p)N$ shots of the ideal state to estimate $\langle \hat{O} \rangle$.

- Finish argument about ideal QEM
- Summary of QEM method types (quick overview) and transfer sentence to VD purity type approach.
- Why VD:
 - operates on noisy physical states rather than requiring logical encoding;
 - is applicable to general Hilbert spaces, i.e. hybrid good
 - exchanges additional copies/sampling/circuit resources for reduced bias with fixed length (given copy and register size) derangement and measure at end
 - is therefore interesting for the multi-register and eventually hybrid architecture considered here.

Error mitigation protocols are diverse in their resource efficiency, assumptions on error-types, and scaling with system size. In this thesis, we consider the virtual distillation protocol [54–56], a QEM method, applicable to both general

Hilbert spaces, including both DV and CV systems, with few assumptions on the underlying noise model [9, 95]. The following section gives details on the mechanism and practical circuits.

2.2.2 Virtual Distillation Protocol

Virtual distillation is a purification-based QEM, utilising k -copies of a circuit to reduce the final bias of the measured expectation value of a noisy state, ρ . The term *purification* is used to signify that the method effectively approximates the pure state closest to ρ , with the underlying assumption that the dominant eigenvector of ρ approximates the ideal pure state, $\rho_{\text{id}} = |\psi_{\text{id}}\rangle\langle\psi_{\text{id}}|$. The validity of this approximation is explored in detail in reference [95], where they derive a bound for the deviation between the ideal state and dominant eigenvector, termed the “coherent mismatch”, c . This error provides the noise-floor for purification based techniques. Favourably, for many error models and circuit types, c is negligible to relevant applications, and in fact scales inversely proportionally to the number of qubits in a single copy, N , for fixed depth circuits.

The purified state may be accessed by,

$$\rho_{\text{pur}}(k) = \frac{\rho^k}{\text{Tr}[\rho^k]}, \quad (2.18)$$

where the denominator acts as a normalisation factor, and for $k = 2$ is the usual definition for the purity of a mixed state. To elucidate the claim that this is the form of the purified state, we expand the spectrum of ρ , assuming that the dominant eigenvector is the ideal state,

$$\rho = \lambda_{\text{id}} |\psi_{\text{id}}\rangle\langle\psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i |\psi_i\rangle\langle\psi_i|, \quad (2.19)$$

with $\lambda_{\text{id}} > \lambda_i$ for all i , and from normalisation, $\lambda_{\text{id}} + \sum_{i=1}^{2^N-1} \lambda_i = 1$. Inserting this expansion into equation 2.18, we find,

$$\rho_{\text{pur}}(k) = \frac{1}{\mathcal{N}} \left(\lambda_{\text{id}}^k |\psi_{\text{id}}\rangle\langle\psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i^k |\psi_i\rangle\langle\psi_i| \right), \quad (2.20)$$

with normalisation factor $\mathcal{N} = \lambda_{\text{id}}^k + \sum_{i=1}^{2^N-1} \lambda_i^k$. The exponentiation acts to relatively attenuate all error eigenvalues compared to λ_{id} , with the limit in k recovering the pure ideal state, $\lim_{k \rightarrow \infty} \rho_{\text{pur}}(k) = |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}|$.

The remaining question is how can we experimentally access $\rho_{\text{pur}}(k)$? In references [55, 56], it is shown that the expectation value of the purified state with respect to the applied operator $\hat{O}$ is,

$$\langle \hat{O} \rangle_{\text{pur}} = \text{Tr}[\hat{O} \rho_{\text{pur}}(k)] = \frac{\text{Tr}[\hat{O} \rho^k]}{\text{Tr}[\rho^k]}, \quad (2.21)$$

and crucially may be estimated without ever physically preparing $\rho_{\text{pur}}(k)$. Instead the numerator and denominator may be found through preparing k -copies of the noisy state, applying some derangement operation to entangle all copies together, and measuring the desired observable on one of the copies. This can be understood through the following identity² providing the numerator,

$$\text{Tr}[\hat{O} \rho^k] = \text{Tr}[\hat{O}^{(i)} \hat{S}^{(k)} \rho^{\otimes k}], \quad (2.22)$$

where $\hat{S}^{(k)}$ is the derangement operator, which permutes the k copies of the state, and $\hat{O}^{(i)}$ is the operator acting on the i th copy. The denominator of the corrected expectation value is found by replacing the operator with the identity operator. There is some freedom in the form of the derangement, here we shall give it explicitly as the cyclic permutation of k objects,

$$\hat{S}^{(k)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_k\rangle = |\psi_2\rangle \otimes |\psi_3\rangle \otimes \dots \otimes |\psi_k\rangle \otimes |\psi_1\rangle. \quad (2.23)$$

Practically, if we can perform this multi-copy derangement, $\hat{S}^{(k)}$, and single-copy operator $\hat{O}^{(i)}$, on k -copies of the noisy state, we can postprocess a collection of circuit repetitions to recover the k th purified expectation value $\langle \hat{O} \rangle_{\text{pur}}$. In the following section we provide two possible approaches for implementing these unitaries and measurements, and give explicit circuit decompositions for both.

Add sample efficiency point

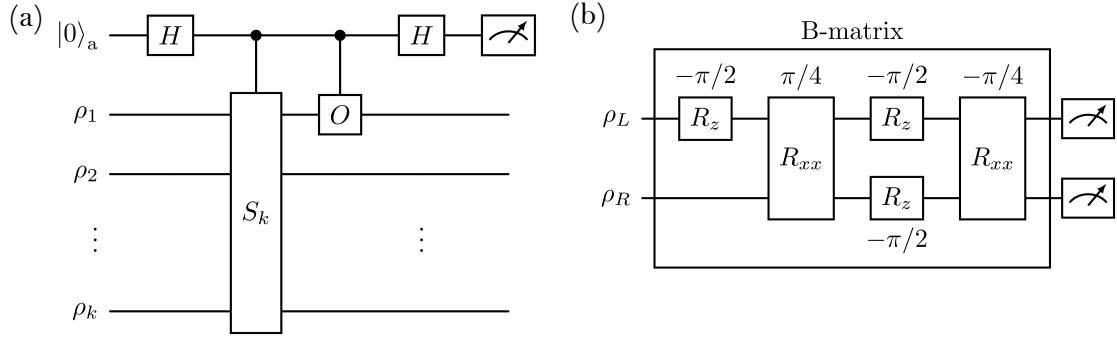


Figure 2.5: Two circuits for implementing virtual distillation. (a) Hadamard test method using controlled derangement, $\hat{S}^{(k)}$, over k copies, and controlled operator, $\hat{O}$, applied on one copy. The ancilla qubit is measured to extract $\text{Tr}(\hat{O}\rho^k)$. Repeating the circuit without the controlled operator measures normalisation factor, $\text{Tr}(\rho^k)$, used to recover the virtually purified result, $\langle\hat{O}\rangle_{\text{pur}}$. (b) Two-copy implementation using a B -circuit, here decomposed into two entangling gates and single qubit phase gates. Measurement of both qubits in the computational basis, and classical postprocessing recovers $\text{Tr}(\hat{Z}\rho^2)$ and purity, $\text{Tr}(\rho^2)$, used to recover $\langle\hat{Z}\rangle_{\text{pur}}$.

Circuits

We discuss two methods of implementing virtual distillation. The first being a general approach able to act on arbitrary Hilbert spaces using Hadamard tests, and the second being more resource efficient, however not simply generalisable to arbitrary sizes of each copy, and to multi-qubit observables.

We begin with the Hadamard test, to build further intuition on virtual distillation based on similar circuit topology discussed in § 2.1.1 for CV-tomography. The circuit for virtual distillation via an ancilla is shown in figure 2.5 (a). The derangement and desired operator, $\hat{O}$, are applied conditioned on the ancilla qubit, with a final measurement only made on the ancilla. The circuit is repeated for N_{shot} times and the outcomes are averaged to recover the estimated real part of the numerator in equation 2.21. The circuit is again repeated N_{shot} times with the controlled operator replaced with the identity operator. Subsequent averaging recovers the denominator, which may then be used to recover $\text{Re}[\langle\hat{O}\rangle_{\text{pur}}]$.

²**Is this footnote on wrong page?** Huggins [56] provides a proof of the identity, which we give an example for the case $k = 2$ in appendix A.9.

The controlled derangement operation may be decomposed into a series of controlled-swap gates [55], which for DV systems may be decomposed to native single and two-qubit gates accessible on the ion trap platform. For CV-systems, the derangement can be decomposed to a series of Fredkin gates [**fredkin_conservative_1982**, **gao_entanglement_2019**], which in turn may be accessible by either controlled-beam-splitters [69, 96] and/or controlled-parity operations [97].

The second method, known as the “ B -matrix” [56], is slightly more resource efficient, not requiring ancilla qubits, however does not simply scale for $k > 2$, and for non- Z measurements. The approach recognises that for $k = 2$ the symmetrised computational observable $\hat{Z}_2 = \frac{1}{2}(\hat{Z}\hat{I} + \hat{I}\hat{Z})$ commutes with the SWAP operator, $\hat{S}^{(2)}$, and consequently so too do both observables required for virtual distillation,

$$[\hat{S}^{(2)}, \hat{Z}_2 \hat{S}^{(2)}] = 0. \quad (2.24)$$

Thus the two observables possess a common eigenbasis and may be simultaneously diagonalised by a single matrix³,

$$\hat{B} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}. \quad (2.25)$$

In the computational eigenbasis, the observable is now given by,

$$\langle \hat{A} \rangle = \text{Tr}[\hat{B}^\dagger \hat{D} \hat{B} \rho], \quad (2.26)$$

$$= \sum_{x \in \{0,1\}^2} d_x p(x), \quad (2.27)$$

where $\hat{A}$ is either $\hat{Z}_2$ or $\hat{S}^{(2)} \hat{Z}_2$, and the second line follows by the definition of the diagonalised operator $\hat{D} = \sum_x d_x |x\rangle \langle x|$, and $p(x) = \langle x | \hat{B} \rho \hat{B}^\dagger | x \rangle$ is the probability of observing computational state $|x\rangle$ after the basis change given by $\hat{B}$.

³Note, we use a different convention than reference [56]. This is done for a simpler decomposition to native gates on our apparatus.

As a result, both numerator and denominator from equation 2.21 may be simultaneously measured by computational basis measurements after applying a two-qubit basis rotation⁴. The values are related to computational expectation values by,

$$\text{Tr}[\hat{Z}\rho^2] = \frac{1}{2}(1 - \langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle + \langle \hat{Z} \otimes \hat{Z} \rangle), \quad (2.28)$$

$$\text{Tr}[\rho^2] = \frac{1}{2}(\langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle). \quad (2.29)$$

Figure 2.5 (b) shows the B -matrix decomposition into native gates to our platform. A demonstration of this simple two-copy example of virtual distillation is given in the following section.

2.2.3 Experimental Demonstration

This section begins with experimental details on implementing virtual distillation via the B -matrix decomposed to native R_{xx} gates and single qubit phase shifts. With this protocol we can then study the efficacy of virtual distillation in two different noise regimes — independent depolarisation, and correlated errors due to common random-unitaries (see § 1.3.3 for the noise model). This work provides evidence of the performance of virtual distillation under realistic structured errors originating from fluctuations of a common external field.

Virtual Distillation on Two Copies

We demonstrate virtual distillation via the B -circuit shown in figure 2.5 (b) on two-copies of a single qubit. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ of a $^{40}\text{Ca}^+$ ion in a quantising field of ~ 5 G. Gates are driven using 729-nm light focussed onto the ions by the individual addressing system detailed in § ???. Two-qubit entangling gates of the form R_{xx} are implemented using Walsh-1 MS gates on the in-phase upper radial mode, as described in § ???characterisation section. For this decomposition, we specifically require $R_{xx}(\frac{\pi}{4})$ -gates. To calibrate these gates, two consecutive $R_{xx}(\frac{\pi}{4})$ pulses are applied, and a partial state-tomography protocol [45] is utilised (as summarised by

⁴Although not discussed here, this method scales simply to Z measurements on copies containing multiple qubits through parallel pairwise rotations [56].

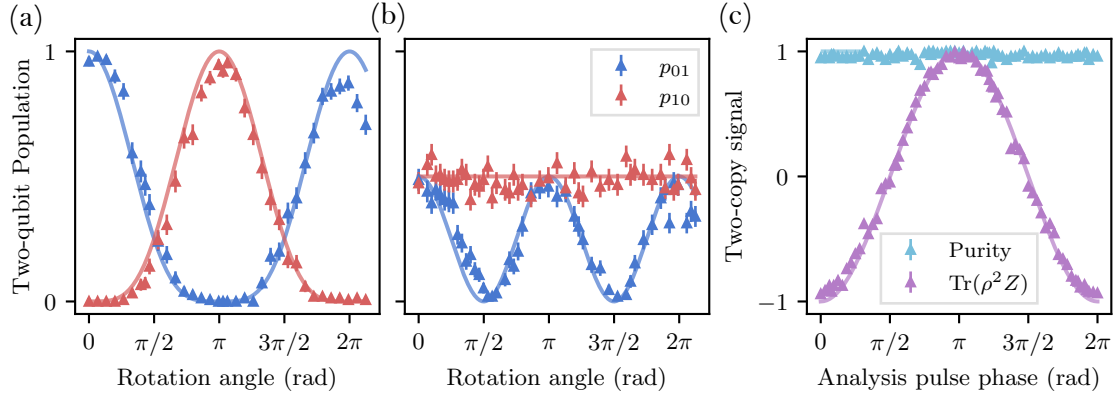


Figure 2.6: Verification of the physically implemented B -circuit. (a-b) Preparing the initial state $|01\rangle$, applying global $R_x(\theta)$ rotations, and measuring the two-qubit joint probabilities provides a clear signal for the correct application of the B -matrix. (a) No B -circuit is applied. (b) B -circuit applied. (c) Globally applied Ramsey experiment followed by the B -circuit and subsequent postprocessing to recover the purity and numerator term given by equations 2.28.

equation ??).

After calibration, the Walsh-1 $R_{xx}(\frac{\pi}{4})$ gate settings are the following: 21 μW total 729-nm beam power, -15 kHz detuning from in-phase upper radial motional mode, 10 μs ramp duration (ramped on and off for both arms of the Walsh), and a 67.5 μs FWHM pulse duration for each arm. The total pulse duration including ramp durations is then 155 μs . Stark-shifts must be carefully calibrated to ensure consecutive MS gates have spin-phases set relative to the qubit. The Stark-shift for this power is calibrated using an SDF on a single ion (described in § ??), finding a shift of 2.49 kHz. This phase is accounted for by updating the RF-tones on the SP AOM for subsequent pulses. With these settings we measure a fidelity for two-consecutive $R_{xx}(\frac{\pi}{4})$ gates of $\mathcal{F}_2 = 96.0(8)\%$, with population component of $p_{01} + p_{10} = 0.010(3)$, and parity contrast of $C = 0.93(2)$. The inferred single $R_{xx}(\frac{\pi}{4})$ gate fidelity, assuming depolarising errors, is then $\mathcal{F} = 98.0(4)\%$.

Single qubit rotations around $\hat{\sigma}_z$ are executed virtually. As we require non-common $\hat{\sigma}_z$ phase shifts on the two qubits, we update the relative phases of the two pairs of RF tones applied to the acousto-optic deflector. As these single-qubit gates are virtual, we expect negligible error contributions.

The final computational-basis measurements are executed by state-dependent

fluorescence, described in § ?? . From the single-shot outcomes, the joint probabilities p_{ij} , with $i, j \in \{0, 1\}$ denoting the bright and dark states of ions 1 and 2 are recovered, respectively. The required expectation values given in equations 2.28 are then simply,

$$\langle Z \otimes I \rangle = p_{00} + p_{01} - p_{10} - p_{11}, \quad (2.30)$$

$$\langle I \otimes Z \rangle = p_{00} - p_{01} + p_{10} - p_{11}, \quad (2.31)$$

$$\langle Z \otimes Z \rangle = p_{00} - p_{01} - p_{10} + p_{11}. \quad (2.32)$$

To initially verify the performance of the physical implementation of the B -circuit, two heuristic tests are used⁵. First, we want a clear indication that the transformation due to the applied B -circuit matches the convention given by the B -matrix in equation 2.25. Inspecting the B -matrix, we see the central block acts distinctly on the $\{|01\rangle, |10\rangle\}$ subspace. As such, a method that preferentially probes this subspace would give a clear signal for conformity between the applied B -circuit and the desired B -matrix. To access this subspace, a singly addressed $R_x(\pi)$ -pulse is applied to prepare the initial state $|01\rangle$. A global $R_x(\theta)$ is then applied with variable rotation angle, θ . Figure 2.6 (a) shows the measured two-qubit populations as a function of θ without the B -circuit applied. Panel (b) shows the populations with the global rotation and the B -circuit applied. The solid lines correspond to the expected populations inferred from equation 2.25, while points are experimental data with $N = 150$ samples per probed rotation angle. We can see good qualitative agreement between the observed data, and the desired behaviour suggesting the phase relationships enforced by the B -circuit indeed conform with the ideal B -matrix.

The second heuristic check verifies the conventions for recovering the numerator and denominator in equations 2.28, from the qubits measured joint probabilities, given in equations 2.30. To test this, a simple Ramsey experiment is performed on both qubits simultaneously. The qubits are initially prepared in $|00\rangle$. A global pulse $R_x(\frac{\pi}{2})$ is then applied, followed by a final analysis pulse $R_\phi(\frac{\pi}{2})$, applied with

⁵Process-tomography would be ideal for this two-qubit implementation, however we have not yet implemented this on our apparatus.

variable spin-phase, ϕ . After these pulses, the B -circuit is applied, and the two-qubit populations are measured. As the two qubits remain in identical pure-states to one another (up to errors in the parallel single-qubit gates), we expect the inferred purity to remain unity, $\text{Tr}[\rho^2] = 1$. Given this, we expect the recovered numerator term to be equivalent to typical Ramsey oscillations on a single qubit,

$$\text{Tr}[\hat{Z}\rho^2] = \langle \hat{Z} \rangle_\phi = -\cos(\phi). \quad (2.33)$$

Figure 2.6 (c) displays the post-processed purity and numerator terms extracted as a function of analysis pulse phase, ϕ , with solid lines showing the expected behaviour of the ideal B -circuit. Again we see excellent conformity between experimental and desired behaviour, giving evidence that we are successfully applying virtual-distillation.

In the next section we use virtual distillation for its intended purpose — to suppress the impact of errors. We will discuss the efficacy of virtual distillation under two different noise models, providing experimental evidence for both.

Virtual Distillation under Correlated Errors

- We target demo of suppression, of controlled injected errors, that are quantified by k-copy RB.
- Give explicit form for suppression of correlated and uncorrelated channels. Look at IRO presentation
- Mention this is found by B circuit applied in Pauli basis and using eqn kcopy pops
- Give motivation of why k-copy RB is particularly relevant to untwirled VD
- Present data in form of single qubit decay rates.
- Give analysis and fits.

To quantify the achieved suppression of errors through the use of virtual distillation, we must characterise and inject errors with a known structure. As such, applying randomised Clifford sequences, the same as those used in standard randomised benchmarking (RB), are a convenient method for error injection. As

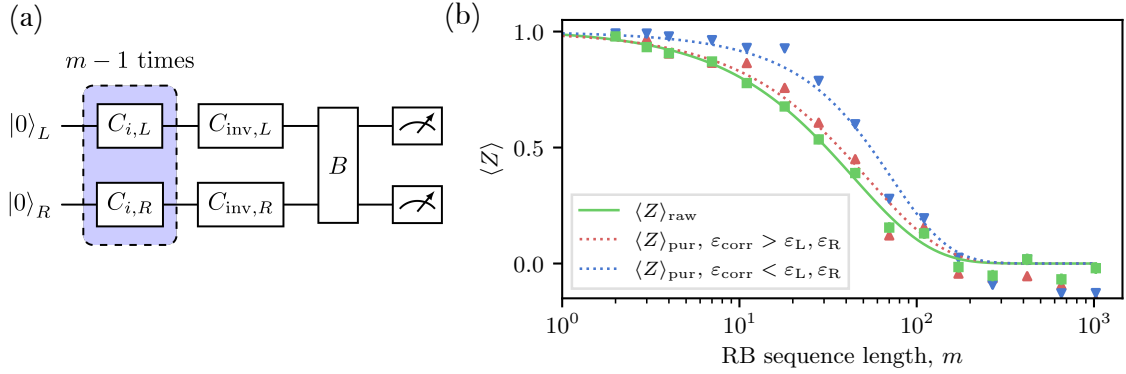


Figure 2.7: Figure: Circuit diagram of RB + VD, exp data for noise floor for two copy B-matrix, with and without spatial correlations

Verifying the performance of virtual distillation under either dominant local depolarising or correlated errors. Structured errors were injected through a fixed frequency detuning of the 729-nm laser, and the application of a sequence of random Clifford gates. In the case that identical Clifford gates, $C_{i,L} = C_{i,R}$, are applied on both qubits, the detuning error is twirled to a *common noise* channel, while in the case that Clifford gates are sampled independently for both copies, the detuning is twirled to local depolarisation channels. The local and correlated injected error-per-Clifford for both regimes are quantified using k -copy RB. Panel (a) shows the circuit of injected errors followed by the B -circuit. Panel (b) shows the effective reduction in error-per-Clifford by applying the virtual distillation circuit. The green points correspond to the raw single-qubit decay due to the physical errors, the solid green line is a fit of this decay to measure a total error-per-Clifford. The red and blue points correspond to the error mitigated Z observables for the dominant correlated errors and local depolarising errors respectively. The dotted lines correspond to the expected error mitigated single qubit decays inferred by the model given in equation ??, and the error-per-Cliffords measured by k -copy RB.

such we frame the task of quantifying error suppression in the context of single-qubit RB by comparing the physical error-per-Clifford, ϵ_{raw} , to the effective error-per-Clifford of the purified observables, ϵ_{pur} .

Figure 2.7 (a) shows the circuit implemented for controlled error injection — random Clifford sequences with a final inverting gate are applied in parallel to both qubit copies, effectively driving depolarising channels with total circuit error rate variable by sequence length. The B -circuit is then applied followed by $\hat{Z}$ measurements. Postprocessing of the two-qubit joint probabilities recover the error-suppressed expectation value, $\langle \hat{Z} \rangle_{\text{pur}}$, and ϵ_{pur} may be inferred by its exponential decay.

The sequence is repeated without the applied B -circuit to recover the standard single-qubit exponential decays, $\langle Z \rangle_{\text{raw}}$ and the corresponding error-per-Clifford,

ε_{raw} .

Crucially, with this circuit we may explore two different error regimes: first, we may apply independent single-qubit Clifford sequences to each copy, which independently twirls each copy, with the aim of decorrelating errors between the two copies [**Clifford twirling**]. Decorrelating errors between qubits ensures the dominant eigenvector of the noisy state is well approximates the ideal pure state, ρ_{id} .

In contrast, the second regime we consider purposefully applying the same Clifford sequence to each qubit, i.e. by enforcing $C_{i,L} = C_{i,R}$. In this regime, any coherent errors acting globally on both qubits will not be decorrelated. These errors cause common drifts of the dominant eigenvector away from the ideal pure state, and as such significantly degrade the performance of virtual distillation. Correlated errors arising from global coherent noise is studied in the previous chapter. Applying identical parallel Clifford gates simplifies the structure of the error channel, but retains these adverse correlations. Such correlated errors are commonplace in the ion-trap platform due to noisy global control fields interacting with the ions. Under parallel Clifford twirling, these global control field errors appear as the random common unitary channel, discussed in § 1.3.3. As such, the relative proportions of uncorrelated, depolarising type errors, to correlated *common noise* errors is required for predicting the performance of virtual distillation. The task of quantifying these errors is addressed in § 1, and we shall use the developed protocol here.

correlated error channel with VD, do this in words? already done?

-> need to show we dont get suppression

Then give analytic form for RB type experiment decomposed to error-per-cliffords

Next give description of experimental data, using detuning to add correlated error, and extracted error per clifford from k-copy RB

Discuss that even in the case of independent Clifford twirling, we still measure a substantial correlated error.

Discuss extracted e-raw and e-pur and the suppression factor
Summarise section, leaning on importance of local twirling.

2.2.4 Outlook

- Summary of what we have shown experimentally so far.
- Linker sentence to next experimental steps.

Extending to > 2 copies

- plan for controlled swap implementation, requirements for this.
 - How many gates, what fidelity we need to ensure tot-err < 1 for derangement side.
 - Given 98% how many copies could we go to? Can we estimate shot numbers here?
- Summary in a table would be nice.
- copy number | derangement gates | circuit fidelity | expected sampling overhead

Virtual Distillation on Hybrid Systems

- Why important to develop, applications in simulations of QO demo, Feynman, gauge theories.
- Cite Quantum virtual cooling
- How it looks on a thermal state and on various quantum states (sqz, coherent, fock)
- How we can implement c-swap using CV gate set.

Limitations

- Never actually remove entropy from system, so we are limited to total error rate < 1 , unlike QEC. This places us firmly in NISQ applications.
- Coherent and structured errors. Mitigation is twirling.
- Coherent noise floor

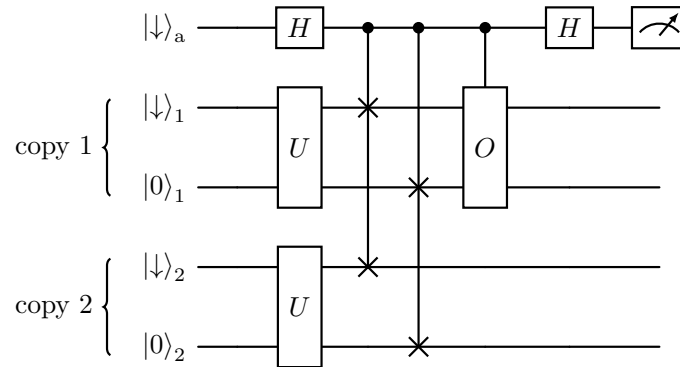


Figure 2.8: Figure: Fredkin gate on oscillator, and how to compose derangement transversally. Proposed circuit for applying virtual distillation to a hybrid system. Using the ancilla method for measuring $\text{Tr}(\hat{O}\rho^k)$, we may decompose the controlled derangement to the transversal application of Fredkin gates.

- Slow readout if mapping to single ancilla?

2.3 Conclusions

- Summary of chapter, saying this gives our future directions.
 - Future planned hardware extensions will give more options for CV.
- Standing wave extension

Appendices

A

Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Gate Fidelity

We derive the fidelity of a noisy gate implementation with respect to the ideal unitary.

A quantum channel,

$$\mathcal{E} : \rho \mapsto \rho, \tag{A.1}$$

which executes ideal unitary, U , with some error, has an average operation fidelity given by,

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \tag{A.2}$$

where the integral is taken over the uniform Haar measure on the space of pure states [98, 99]. We shall see that the language used for this analysis of single qubit

gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right), \quad (\text{A.3})$$

where V is the error unitary, and d is the dimension of the Hilbert space.

Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.2 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2, \end{aligned} \quad (\text{A.4})$$

where $W = U^\dagger V$. Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.5})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.6})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These

two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.7})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)}(\mathbb{I} + S). \quad (\text{A.8})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.9})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter ???. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.3.

A.6 Projectors of $Cl_1^{\otimes 2}$

We outline the method of recovering the α parameters described in section 1.3.3 using projectors onto the irrep spaces. A special thanks to Iver Øvergaard who performed this analysis applied to the *common noise* channel and gave permission for reproducing the results here **IRO please check you are happy with me to include this appendix.**

The decay due to a single rotation acting identically on both qubits which, in the Pauli-transfer matrix (PTM) representation, can be written as $\Lambda^{\mathcal{P}} = \text{diag}(1, R, R, R \otimes R)$, where superscript $\mathcal{P}$ represents the block diagonal matrix is ordered by the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, with $\vec{\sigma} = \{\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$, and

$R = R(\vec{n}, \theta) \in SO(3)$ describes a rotation by θ around $\vec{n}$. Finding the vector $\vec{\alpha}$ can now be done using Schur's lemma for the case of no mixing between irrep spaces:

$$\tilde{\Lambda}^{\mathcal{P}} = \sum_{\lambda} \alpha_{\lambda} \Pi_{\lambda}, \quad \alpha_{\lambda} = \frac{1}{d_{\lambda}} \text{Tr}(\Pi_{\lambda} \Lambda^{\mathcal{P}}). \quad (\text{A.10})$$

Here, Π_{λ} denotes the PTM projector onto the irrep space labelled λ and with dimension d_{λ} . It follows immediately that the identity sector is fixed, while the left- and right-qubit sectors produce $\alpha_L = \alpha_R = \text{Tr}(R)/3$. In principle, α_{λ} for the two-qubit sector can be evaluated by explicitly constructing the projectors from the basis operators in Table 1.1. However, in this case the symmetries of these irrep spaces allow the projectors to be expressed more simply in terms of operators whose traces are easily evaluated:

$$\Pi_{tr} = T, \quad (\text{A.11})$$

$$\Pi_{dt} = D - T, \quad (\text{A.12})$$

$$\Pi_{so} = \frac{1}{2}(I_9 + S) - D, \quad (\text{A.13})$$

$$\Pi_{as} = \frac{1}{2}(I_9 - S), \quad (\text{A.14})$$

where $T = 1/3 \sum_{ij} |ii\rangle \langle jj|$, $D = \sum_i |ii\rangle \langle ii|$, and $S = \sum_{ij} |ji\rangle \langle ij|$ are the trace projector, diagonal projector, and swap operator, respectively. The required traces are

$$\text{Tr}(I_3 R) = \text{Tr}(R) = (1 + 2c), \quad (\text{A.15})$$

$$\text{Tr}(I_9(R \otimes R)) = (\text{Tr}(R))^2 = (1 + 2c)^2, \quad (\text{A.16})$$

$$\text{Tr}(T(R \otimes R)) = \frac{1}{3} \text{Tr}(R^T R) = 1, \quad (\text{A.17})$$

$$\text{Tr}(D(R \otimes R)) = \sum_i R_{ii}^2 = c^2 + 2c + (1 - c)^2 \kappa, \quad (\text{A.18})$$

$$\text{Tr}(S(R \otimes R)) = \text{Tr}(R^2) = 4c^2 - 1, \quad (\text{A.19})$$

where $c = \cos(\theta)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Substituting these traces into equation A.10

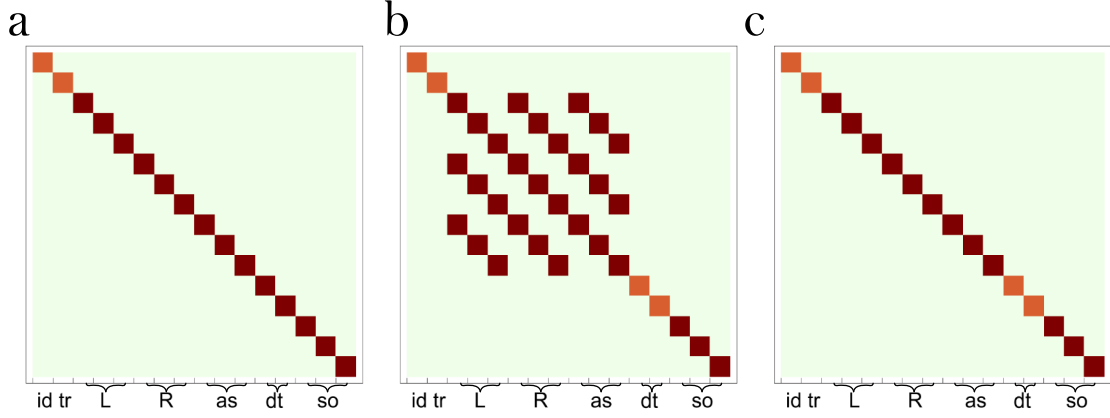


Figure A.1: Liouville operator representation of error channels after twirling over $Cl_1^{\otimes 2}$, ordered by the irrep basis (Schur basis). (a) *Common noise*, (b) *Continuous ZZ*, (c) *Stochastic ZZ*. Orange blocks represent an entry of 1, green represent 0, and red represent a decay with dependence on m .

and using equation A.11 yields

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (\text{A.20})$$

$$\alpha_{tr} = 1, \quad (\text{A.21})$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2\kappa), \quad (\text{A.22})$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2\kappa). \quad (\text{A.23})$$

A.7 ZZ-type noise

In the main text we explore a separable error channel that induces correlations between qubits in a register. Here we shall explore entangling type noise, which although not prevalent on the ion-trap qubit platform, is a dominant error source in superconducting qubits.

The always on ZZ-type error common to superconducting qubits is described by the Hamiltonian,

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z, \quad (\text{A.24})$$

where J_{ZZ} is the coupling strength. Generally, H_{ZZ} will couple the subrepresentations of the 3-irrep, given in table 1.1. We can show the qualitative difference

in the subspace dynamics by plotting the Liouville representation of the twirled *common noise*, and the *ZZ*-type noise. Specifically we plot $\tilde{\Lambda}^m$, where m is the sequence length. Figure 1.5 plots the maps in the Schur basis ordered so isotypic representations are adjacent to one another. Orange blocks have entries of 1, light green blocks are 0, and red blocks represent some decay parameters dependent on m . Panel (a) shows the fully diagonal *common noise* transfer matrix, where we can see the invariant trace subspace that was noted in the main text. Panel (b) shows the general case for twirled *ZZ*-type noise, given by equation A.24. The off-diagonal entries exactly couple the three subspaces of the 3-irrep. We note that the resulting block diagonal form is inconvenient to express as a general multi-exponential decay profile as given in the main text for *common* and local depolarising type noise. However, the resulting block diagonal Liouville form may be used directly with experimental data to fit these parameters of interest. Further, we may still explore the expected asymptotic decays due to this model, and find that under dominant *ZZ*-type noise, we expect an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/2$.

We may recover a simple diagonal Liouville form in the case of stochastic *ZZ* noise, given by,

$$\mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ}(Z \otimes Z)\rho(Z \otimes Z), \quad (\text{A.25})$$

where p_{ZZ} is the probability of a correlated phase flip error on both qubits. In the irrep basis, the following sector decays are found for the stochastic *ZZ* error,

$$\begin{aligned} \alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} &= 1 - 4p_{ZZ}/3, \\ \alpha_{tr} = \alpha_{dt} &= 1. \end{aligned} \quad (\text{A.26})$$

We show this dependence in figure A.1 (c). The same intermediate asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 0.5$ is expected.

The difference in correlated decay structure due to *common*, depolarising, and *ZZ*-type errors suggests that twirling over $Cl_1^{\otimes 2}$ both simplifies a generic error map to be measurable via RB, but also, retains some noise-dependent structure

of the correlated decay profiles. Designing optimal initial and final states, as well as optimal sampling strategies for extracting a larger set of error parameters covering multiple physical error mechanisms is left for future work. A possible strategy to avoid introducing arbitrary error channels, is to be agnostic of the physical error mechanisms, and rather directly extract the $\vec{\alpha}$ and $\vec{m}$ parameters described in section 1.3.2. Analogous protocols to this is described by character RB [36], and tomography via RB [100].

A.8 Magnus expansion

The unitary evolution due to a time dependent Hamiltonian may be written in the ordered, short-time limit as,

$$U(t) = \lim_{N \rightarrow \infty} e^{\frac{-i}{\hbar} \hat{H}(t_N) \Delta t} \dots e^{\frac{-i}{\hbar} \hat{H}(t_2) \Delta t} e^{\frac{-i}{\hbar} \hat{H}(t_1) \Delta t} = e^{A(t)}, \quad (\text{A.27})$$

where $\Delta t = t/N$, and $t_n = n\Delta t$, and $A(t)$ is a time dependent exponent, which is expanded in the Magnus construction to [82],

$$A(t) = \sum_{k=1}^{\infty} A_k(t). \quad (\text{A.28})$$

If the Hamiltonian at different times does not commute, then the first three $A_k(t)$ terms are given by,

$$A_1(t) = \frac{-i}{\hbar} \int_0^t dt_1 \hat{H}(t_1) \quad (\text{A.29})$$

$$A_2(t) = -\frac{1}{2\hbar^2} \int_0^t \int_0^{t_1} dt_1 dt_2 [\hat{H}(t_1), \hat{H}(t_2)] \quad (\text{A.30})$$

$$A_3(t) = \frac{i}{6\hbar^3} \int_0^t \int_0^{t_1} \int_0^{t_2} dt_1 dt_2 dt_3 \left([\hat{H}(t_1), [\hat{H}(t_2), \hat{H}(t_3)]] + [\hat{H}(t_3), [\hat{H}(t_2), \hat{H}(t_1)]] \right). \quad (\text{A.31})$$

This may be understood as the continuous analogy of the Baker-Campbell-Hausdorff formula. It is particularly useful in this thesis to understand the origin of the geometric phase in the MS gate scheme, and the non-linear bosonic interactions generated in the “two-SDF” protocol given in § 2.1.1.

A.8.1 “Two-SDF” dynamics

Here we analyse the “two-SDF” Hamiltonian given in equation A.32 using the Magnus expansion. We recover effective Hamiltonians for both Gaussian interactions, as well as for tri-squeezing.

The “two-SDF” Hamiltonian is reproduced for convenience here,

$$\hat{H}_{2\text{SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (\text{A.32})$$

$$= \frac{\hbar\Omega_\alpha}{2} \hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (\text{A.33})$$

$$+ \frac{\hbar\Omega_\beta}{2} \hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right). \quad (\text{A.34})$$

The first three terms of the corresponding Magnus expansion are found by inserting $H_{2\text{SDF}}$ into equations A.29. We shall later rationalise that with certain values of SDF magnitudes, detunings, and phases, we may arbitrarily suppress the higher order terms.

The A_1 term appears as separable fast oscillations in phase space due to each SDF,

$$A_1(t) = -i \sum_{i \in \{\alpha, \beta\}} \left(\frac{\Omega_i \hat{\sigma}_i}{\Delta_i} \sin\left(\frac{\Delta_i t}{2}\right) \left(\hat{a}_i e^{i\left(\frac{\Delta_i t}{2} + \phi_i\right)} + \hat{a}_i^\dagger e^{-i\left(\frac{\Delta_i t}{2} + \phi_i\right)} \right) \right). \quad (\text{A.35})$$

This is effectively the first-order only dynamics of the displacement operator leading to excursions in phase-space. The effects of this term may be entirely suppressed at certain gate durations, $A_1(T) = 0$, where T is an integer multiple of both $\tau_\alpha = 2\pi/\Delta_\alpha$ and $\tau_\beta = 2\pi/\Delta_\beta$. Excursions may also be made arbitrarily small by ensuring $\Omega_i \ll \Delta_i$.

The commutator in A_2 term may be expanded to reveal four relevant components,

$$A_2(t) = S_\alpha(t) + S_\beta(t) + X_{\text{pair}}(t) + X_{\text{exch}}(t), \quad (\text{A.36})$$

where S denotes “self-terms”, and X “cross-terms” of the two simultaneously applied SDFs. The self-terms, with the form,

$$S_i(t) = \frac{i\Omega_i^2}{4} \left(\frac{\sin(\Delta_i t)}{\Delta_i^2} - \frac{t}{\Delta_i} \right) \quad (\text{A.37})$$

are recognisable from section ?? as geometric phase factors. Taking the assumption that $t \gg \frac{1}{\Delta_i}$, we get a linearly growing geometric phase term, $S_i(t) \approx \frac{-i\Omega_i^2 t}{4\Delta_i}$. In the

case that this geometric phase acts globally on the system, they may be ignored. If the accumulated phase is spin, or oscillator state dependent, then interesting dynamics may be observed, however these effects are out of the scope of this thesis. The remaining cross-terms contain the 2nd order non-linear bosonic interactions we are interested in emulating. To recover the general family of non-linear bosonic interactions given in table 2.1, motional phases are set to $\phi_\alpha = 0$, $\phi_\beta = \phi$, and spin-conditioning is assumed to be the anti-commuting pair: $\sigma_\alpha = \sigma_x$, $\sigma_\beta = \sigma_y$. We set the detunings of the two SDF to be $\Delta_\beta = m\Delta_\alpha$, where m is some integer. Expanding the commutator of the cross terms we find,

$$X_{\text{pair}} =, \quad (\text{A.38})$$

$$X_{\text{exch}} =, \quad (\text{A.39})$$

$$(\text{A.40})$$

where... We see now the freedom to “tune” the desired interaction, if we set $m = +1$, we resonantly drive..., alternatively if we set $m = -1$, we recover... We can convert from the generator form, $A(T)$, to familiar effective time-independent Hamiltonian, H_{eff} for a given time, T , by $\hat{H}_{\text{eff}} = \frac{i\hbar}{T}A(T)$.

For $m = +1$, we find that the effective Hamiltonian is,

$$H_{\text{eff,exch}} = \dots \quad (\text{A.41})$$

which for a single mode, $a^\dagger = b^\dagger$, is the parity operation, and for two modes, is the beam splitter.

For $m = -1$, we find,

$$H_{\text{eff,pair}} = \dots \quad (\text{A.42})$$

which for a single mode, $a^\dagger = b^\dagger$, is the squeezing operation, and for two modes, is the two-mode squeezing.

Similar arguments hold for $m = 2$, where now the A_2 cross terms appear off-resonant by..., and the on resonant cross terms in A_3 coherently sum over time yielding effective 3rd order bosonic operators, such as the tri-squeezing interaction.

Higher order error term scaling...

A.9 Derangement for $k = 2$

Huggins [56] provides a proof of the identity given in equation 2.23, which I reproduce here for the simple example of $k = 2$. The derangement operator for two copies is simply the swap operator, $S^{(2)} = \text{SWAP}$, and we will assume the observable acts only on the first copy, giving $\hat{O}^1 = \hat{O} \otimes \mathbb{I}$. The identity is then,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (\text{A.43})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \langle i_1 | \hat{O} | k_1 \rangle \delta_{i_2, k_2} \langle k_1, k_2 | S^{(2)} | j_1, j_2 \rangle \langle j_1 | \rho | i_1 \rangle \langle j_2 | \rho | i_2 \rangle \quad (\text{A.44})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \hat{O}_{i_1, k_1} \delta_{i_2, k_2} \delta_{k_1, j_2} \delta_{k_2, j_1} \rho_{j_1, i_1} \rho_{j_2, i_2} \quad (\text{A.45})$$

and contracting the Kronecker deltas, we find,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (\text{A.46})$$

$$= \sum_{i_1, j_1, j_1} \hat{O}_{i_1, j_2} \rho_{j_1, i_1} \rho_{j_2, j_1} \quad (\text{A.47})$$

$$= \text{Tr}[\hat{O} \rho^2], \quad (\text{A.48})$$

with the last equality following from the definition of matrix multiplication.

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